From routine cine CMR to regional myocardial dysfunction:

Fidelity, validation and interpretation of motion and strain estimates

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Abstract

Regional myocardial dysfunction can be clinically important despite preserved global ventricular function. Routine cine cardiovascular magnetic resonance (CMR) permits retrospective motion and strain estimation, but its changing image intensities and boundaries do not explicitly label myocardial material points. Regional interpretation therefore depends on the estimator's assumptions and on what its validation actually measures.

This critical narrative review compares feature tracking, registration, biomechanical and learning-based methods by their inputs, representations, priors and supervision. It separates evidence from known-motion tests, tagging or displacement encoding with stimulated echoes (DENSE), repeatability, tissue characterisation and clinical outcomes. Selected paired studies support circumferential-strain agreement at whole-slice, global and segmental scales under matched definitions. DENSE-supervised learning improved end-systolic cine–DENSE agreement, with residual individual differences and no external-dataset validation. Radial measurements are particularly sensitive to product and definition. A prescribed-scar evaluation of DeepStrain on MRXCAT2.0 reported component-specific errors averaged over whole slices; those results do not quantify recovery within the scar. Clinical discrimination and prognosis add useful evidence, but do not establish material-motion accuracy.

A defensible regional claim should specify the acquisition, estimator, population, spatial scale, strain definition, reference and tolerance. Controlled known-motion tests and aligned material-sensitive in-vivo comparisons can probe abnormality magnitude, location, extent and timing; downstream studies address clinical relevance. Input information, representation, estimation and reference uncertainty can contribute jointly to error. Matching the claim to these complementary tests is more informative than treating any single agreement measure as general validation.

Keywords: cardiovascular magnetic resonance; cine CMR; myocardial motion; myocardial strain; regional dysfunction; feature tracking; image registration; validation.

Plain English summary

The heart can contain a small area that moves abnormally even when its overall pumping measure looks normal. Standard cine CMR provides moving pictures of the heart and is available in routine care. Computer methods can use these pictures to estimate motion and strain, a measure of deformation. An

ordinary cine image, however, does not label each piece of heart muscle, so different methods can produce different internal motion while matching the same heart borders.

We reviewed how cine-based methods work and how they have been tested. Some studies show useful agreement with specialised CMR methods that are more sensitive to tissue motion. Other studies show important differences between software products, strain directions and heart regions. A good-looking contour, a repeatable number, an association with scar and a prediction of outcome are all valuable findings, and each of them answers a different question.

Regional measurements should therefore be judged for the exact property they are meant to report: the severity, position, extent or timing of an abnormality. Tests with known motion should be combined with carefully registered comparisons to specialised scans in people. The method, scan settings, strain definition and acceptable error should be stated in advance. This approach can identify where routine cine CMR gives reliable regional information and where the result should remain qualified.

1 Why regional function is a distinct measurement problem

Left-ventricular (LV) ejection fraction compresses a complex three-dimensional deformation into a chamber-level ratio. That compression has real value: ventricular volumes and ejection fraction are reproducible CMR endpoints, widely understood and embedded in clinical reporting standards [1, 2]. The same compression also hides information. Well-functioning regions can compensate for a smaller abnormal region, opposing changes can cancel in an average, and the timing of regional contraction can change without a large fall in ejection fraction. Regional motion and strain are attractive precisely in the clinical settings where such hidden changes matter: focal infarction, heterogeneous cardiomyopathy, dyssynchrony, early dysfunction and treatment response.

Clinical studies support the relevance of measuring deformation in those settings. Feature-tracking strain has been associated with functional recovery after myocardial infarction, adverse remodelling and cardiovascular outcomes [3–5], and DENSE circumferential strain has shown prognostic association after acute infarction [6]. These associations motivate evaluating deformation. They leave open how accurately a regional abnormality has been located or quantified, because a prognostic marker can remain useful despite measurement bias, and a technically accurate motion field may still fail to improve a clinical decision. Analytical validity, biological association and clinical utility are linked questions with separate answers [7, 8].

Earlier reviews have covered the analytical-validity question at the level of strain physics, modality differences, clinical practice and sources of discrepancy [9, 10], and more recent reviews cover deep learning for wall-motion analysis across modalities and strain from CMR and computed tomography [11, 12]. The present review addresses a narrower question within that literature. It asks how faithfully routine cine CMR can recover *regional or focal* myocardial dysfunction, and how the choice of validation target changes the claim that can be made. Answering that question requires comparing estimator families by their inputs, assumptions, estimands, populations and reference standards: how each family uses image information, representation, priors and supervision to infer motion and strain, and which spatial and temporal scales of measurement its validation evidence supports. Within that comparison, the focal case is the most demanding one, and it is used here as a test of the evidence base, without treating every method paper as if it had set out to solve it.

The practical question is:

> For a specified cine acquisition, estimator and measurement definition, which attributes of a regional abnormality (magnitude, location, spatial extent and timing) are recovered within a

tolerance that is appropriate for the intended use?

This formulation steers between two unhelpful extremes. At one extreme, routine cine would be treated as information-free; in reality myocardial borders, papillary muscles, trabeculation, intensity variation, neighbouring structures and temporal consistency all constrain motion. At the other extreme, a visually plausible deformation or an accurate contour would be treated as proof of correspondence; in reality neither uniquely identifies how material points moved within the myocardium. Between these extremes, the evidence supports bounded successes, and the review is organised to state those bounds.

1.1 Review approach and scope

This critical narrative review asks what routine cine CMR can establish about regional myocardial motion and strain. It combines selected method papers, empirical comparisons, reviews and public-resource descriptions; it is not a systematic review or a census of cine-strain studies. The search record describes citation expansion from a core set and a structured search of PubMed, Europe PMC and OpenAlex executed on 3 September 2026, with a publication cutoff of 1 September 2026 and a cap of 100 records per query. The six query families concerned regional strain after infarction, known-motion validation, prior perturbation, conditional recoverability, failure-aware prediction and regional localisation or attenuation. The surviving query log records 18 query–database combinations; the original record exports and record-level screening file are not available for re-audit.

The historical notes describe a single software-assisted title-and-abstract screening pass, followed by a directed check on 9 September under the same cutoff. Their deduplication, retention counts and thematic stopping rule cannot be reconstructed from the surviving records. A separate October update added or rechecked sources, including publications after the September cutoff. On 11 October 2026, central quantitative claims were rechecked against primary-source methods, results and tables, and their interpretation was corrected where needed. This targeted verification is documented in the Supplementary Methods; it does not reconstruct the historical search or establish exhaustive coverage.

Reading depth varies by source. Central comparisons were checked in the relevant full-text sections where accessible; other method-lineage entries rely on abstracts or bibliographic records and support only the information available at that depth. The updated evidence tables distinguish those readings from inherited extraction notes and failed access. Missing detail in a retrieved abstract is not evidence that the original study omitted an experiment. Accordingly, the review identifies priorities for focal validation without claiming that a design is absent from the entire literature. Historical counts and the limits of their auditability are retained in the Supplementary Methods.

The studies discussed in Sections 3 to 6 were compared along six axes: (1) image input and sampling; (2) population and abnormality; (3) estimated entity, such as contour motion or dense displacement; (4) strain definition and aggregation; (5) validation target; and (6) reported error or association. Because these axes, reference standards and endpoints were too heterogeneous to share a scale, results were not pooled. Dice overlap, displacement error, strain bias, scan–rescan precision, scar discrimination and outcome association were kept as different quantities throughout.

2 From cine images to a regional measurement

2.1 What routine cine CMR observes

Balanced steady-state free-precession (bSSFP) cine CMR samples a time series of image intensities. In a typical short-axis image, the blood–myocardium interfaces are conspicuous, whereas intramyocardial texture can be weak and changes with coil sensitivity, through-plane motion, partial volume, reconstruction

and artefact. The image therefore supplies anatomical and intensity cues without a persistent identity label for every tissue element. Every estimator has to add something to those cues. Feature tracking exploits image features or propagated contours; registration exploits an image-similarity relation; biomechanical models constrain admissible deformation; and learning-based methods import associations learned from training data [13, 14]. These four categories overlap in practice: a neural network may predict registration parameters, a registration method may use a learned similarity measure, and a biomechanical penalty may be embedded in either.

The separation between observed cues and inferred correspondence in Figure 1a is a causal one, and it can be written down. Let a point in a reference configuration be $\boldsymbol{X}$, and let an estimated mapping be $\boldsymbol{\varphi}(\boldsymbol{X}, t)$. The displacement and deformation gradient are

$$\boldsymbol{u}(\boldsymbol{X}, t) = \boldsymbol{\varphi}(\boldsymbol{X}, t) - \boldsymbol{X}, \qquad \boldsymbol{F}(\boldsymbol{X}, t) = \frac{\partial \boldsymbol{\varphi}}{\partial \boldsymbol{X}} = \boldsymbol{I} + \frac{\partial \boldsymbol{u}}{\partial \boldsymbol{X}}. \tag{1}$$

One possible finite-strain measure is the Green–Lagrange tensor,

$$\boldsymbol{E} = \frac{1}{2}(\boldsymbol{F}^{\mathsf{T}}\boldsymbol{F} - \boldsymbol{I}). \tag{2}$$

These equations (Equation (1) and Equation (2)) state how a chosen mapping is converted into a chosen strain; they say nothing about whether $\boldsymbol{\varphi}$ is correct. The conversion itself has consequences for error. Differentiation can amplify local mapping error, while smoothing can reduce variance and alter peaks or spatial gradients. The reported number further depends on coordinate axes, myocardial layer, reference frame, segment definition, spatial aggregation and whether "peak" means the minimum of a segment-average curve or the average of pointwise minima. A comparison that ignores these choices can attribute a definitional difference to a tracking error [24, 25].

2.2 The same contours can support different material mappings

Boundary agreement constrains deformation without determining it. Consider an annulus whose end-diastolic and end-systolic contours are fixed. Material points can retain their angular positions, or they can be reparameterised around the same final contour. Both mappings give Dice overlap of one and Hausdorff distance of zero for the masks, while their circumferential stretch varies differently with position (Figure 2). The counterexample proves a limited point: equal masks or equal mean geometry are insufficient to identify intramyocardial correspondence. It leaves two other questions open, namely whether cine-derived regional strain is wrong in any particular case, and whether image texture and temporal information can select a useful mapping.

The consequence of this non-uniqueness grows as the claim becomes more local. Chamber volume can be estimated from boundaries alone. A global contour-length change is also strongly boundary-constrained. A segmental average additionally requires correspondence between sectors. Transmural or voxelwise strain requires, on top of that, intramyocardial correspondence and spatial derivatives. Along this sequence, a method can be accurate for volumes and useful for global strain while remaining insufficiently validated for focal strain gradients. Validation should therefore follow the finest claimed spatial and temporal scale; credibility earned at a coarser scale does not transfer downward.

2.3 A complete estimand is more than the word "strain"

Because each step of that sequence adds a definitional choice, a regional strain estimand should state at minimum:

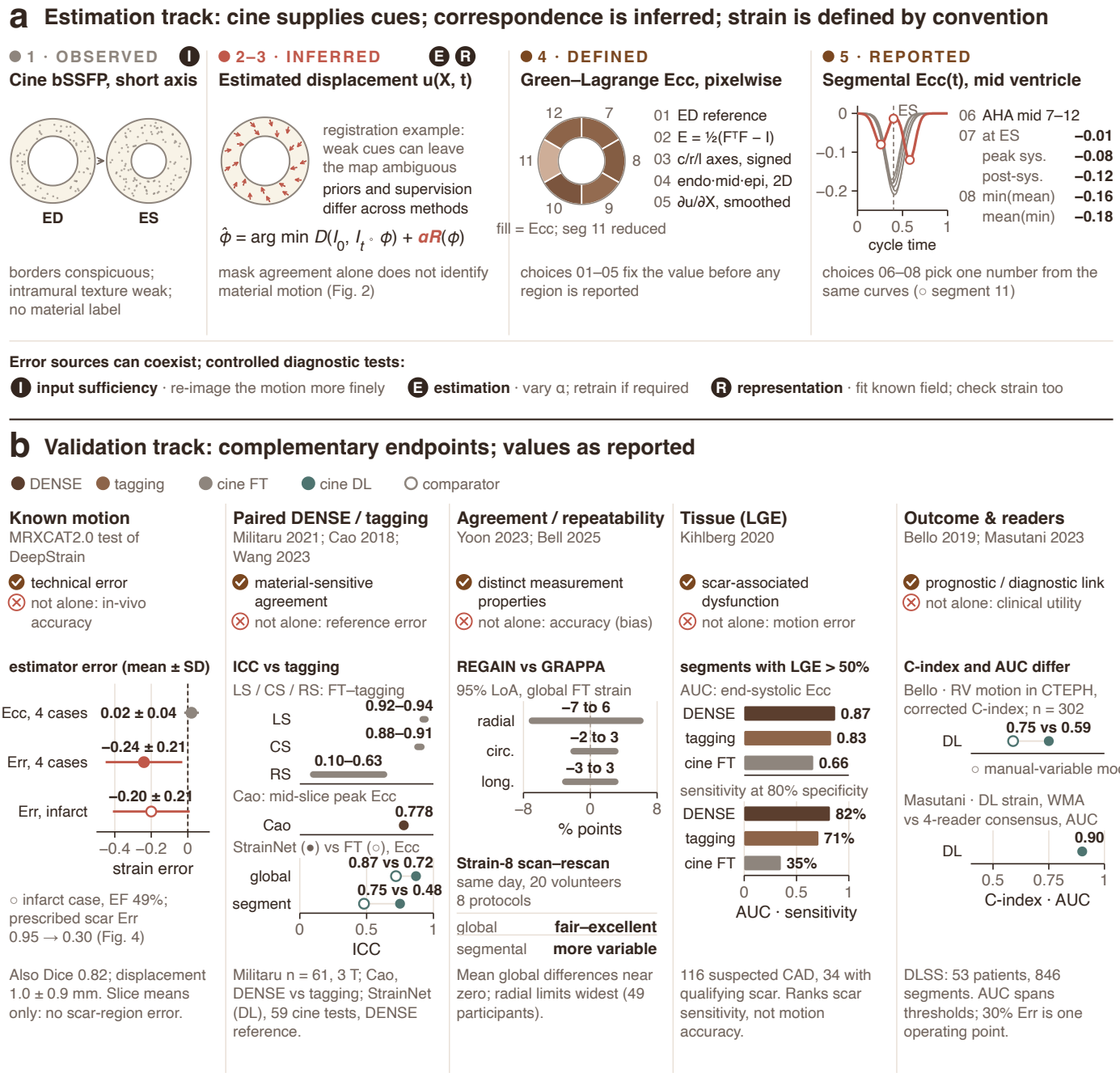


Figure 1. From image cues to a regional measurement. (a) Conceptual chain: correspondence is inferred and strain depends on the measurement definition. The registration objective shown is one example; other families optimise other targets. Segment 11 is reduced; all curve readouts are calculated illustrations. Input (I), estimation (E) and representation (R) tests probe potentially coupled errors. (b) Selected empirical endpoints, shown without pooling or ranking. MRXCAT2.0 reports slice averages [15]. Militaru intraclass correlation coefficients (ICCs) for longitudinal, circumferential and radial strain (LS, CS, RS) compare feature tracking (FT) with tagging; the global RS range is 0.10–0.63 [16]. Cao reports mid-slice peak circumferential strain (Ecc) [17]; StrainNet reports end-systolic cine–DENSE Ecc agreement [18]. REGAIN assesses reconstruction-plus-analysis agreement [19]; Strain-8 assesses repeatability [20]. Kihlberg's receiver operating characteristic (ROC) analysis uses end-systolic segmental Ecc and late gadolinium enhancement (LGE) [21]. Bello's corrected C index concerns right-ventricular (RV) survival prediction in chronic thromboembolic pulmonary hypertension (CTEPH) [22]; the deep-learning synthetic strain (DLSS) area under the curve (AUC) evaluates reader-defined wall-motion abnormality across thresholds [23]. The axis value 0.5 is a scale reference only; no result is plotted there.

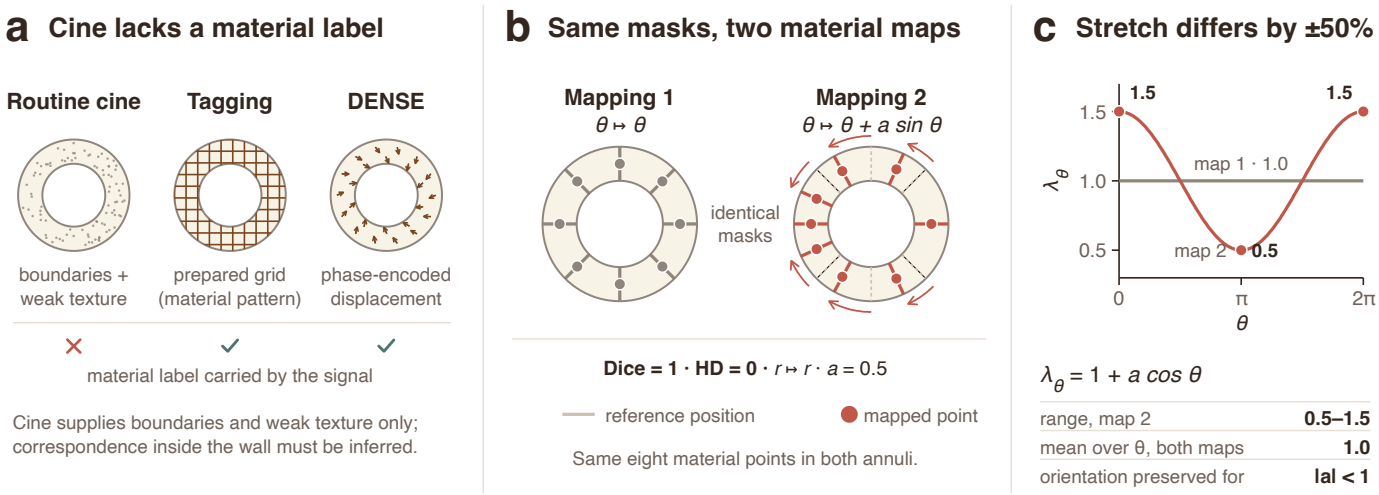


Figure 2. Identical contours do not determine the same material correspondence. (a) Schematic of the information each acquisition supplies: routine cine provides boundaries and weak texture, tagging a prepared pattern, DENSE phase-encoded displacement; only tagging and DENSE carry a material label in the signal. (b) Two material mappings between the same annular contours ($r \mapsto r$): angular identity, $\theta \mapsto \theta$, and $\theta \mapsto \theta + a \sin\theta$ with $a = 0.5$. Both give identical masks (Dice = 1, Hausdorff = 0). (c) Their circumferential stretch, $\lambda_\theta = 1$ and $\lambda_\theta = 1 + a\cos\theta$, differs; the latter ranges from 0.5 to 1.5 and preserves orientation for $|a| < 1$, while both mappings have the same mean stretch over θ (1.0). The figure is an analytic counterexample; no estimator error is measured, and the stretch ratio is a different quantity from Green–Lagrange strain.

- the reference configuration and mapping direction;
- the finite-strain measure (for example, linear, Green–Lagrange or logarithmic strain);
- the anatomical coordinate system and sign convention;
- the myocardial layer and dimensional model (2D slice, 2D+time or 3D+time);
- the differentiation, interpolation and spatial regularisation used;
- the region definition, registration to the American Heart Association (AHA) segment model and treatment of junctions;
- the temporal summary, including end-systolic versus peak strain and handling of post-systolic shortening; and
- the order of spatial and temporal aggregation.

When these items are missing, apparent disagreement may reflect different quantities. The problem is most prominent for radial strain, for which wall-thickening surrogates, tensor components and layer-specific measurements are not interchangeable. It also affects circumferential comparisons whenever one pipeline reports one-dimensional engineering strain and another reports a component of a two-dimensional Green–Lagrange tensor [17, 25].

3 Estimator families and the assumptions they contribute

All estimators resolve incomplete image information by adding structure. The useful comparison between them therefore concerns which assumptions each introduces, whether those assumptions are visible to the user, and whether they suit the abnormality being studied. Table 1 summarises these attribute-level distinctions. Section 3.1 then uses registration to explain how information, representation and estimation interact, and Section 3.2 to Section 3.6 follow the literature in approximate chronological order within

each family. Supplementary Table S3 and Figure 3 map selected method and evaluation entries, including non-cine comparators. Their documented endpoints do not provide a field-wide prevalence estimate.

Table 1. Method attributes that may coexist within a cine-CMR motion estimator. Rows are descriptive archetypes; a single method may combine several of them.

Archetype	Input	Output and estimand	Motion representation	Constraint/prior	Optimisation or training target	Endpoint directly validated in representative work
Feature and contour tracking	Cine pixels plus initial contours	Contour/mesh motion; product-defined strain	Boundary points, meshes or proprietary field	Feature similarity and trajectory continuity	Frame-to-frame match or contour propagation	Global/segmental peak strain, repeatability and tissue discrimination; product-specific
Deformable registration	Image pairs or cine sequence	Dense displacement; derived strain	Voxel field, spline, spectral or implicit field	Spatial and temporal regularity; optional topology	Image/feature similarity plus penalties	Image/shape alignment; selected phantom, tagging or DENSE comparisons
Finite-element (FE) model	Images, masks or meshes	Nodal displacement and tensor strain	Finite-element or reduced-order basis	Compatibility, volume constraint, constitutive/fibre terms	Image correlation plus mechanical residual	Landmark/curve or phantom error in the specified model domain
Supervised learning	Cine or contours plus motion/strain labels	Learned displacement or strain	Dense field or parameter map	Architecture, augmentation and label domain	Label loss; optional image, shape or mechanics terms	Depends on label: StrainNet separates DENSE-contour displacement error from cine-contour Ecc
Self-supervised learning	Cine without motion labels	Dense displacement/strain	Usually voxel or multi-scale field	Smoothness, cycle, topology or shape	Image, feature or mask agreement	Commonly registration and anatomy; material endpoints require separate testing
Compact basis	Any compatible image estimator	Coefficients decoded to motion/strain	Fourier, spline, temporal or neural basis	Band limit or basis dimension	Inherits estimator objective; may be learned	Capacity and final estimator performance are distinct endpoints

3.1 An underdetermined problem, and what the prior is doing

Let $I_t(\boldsymbol{x})$ denote image intensity at position $\boldsymbol{x}$ in phase t, and let $\boldsymbol{\varphi}_{0\to t}$ map a material point in the reference configuration to its position at t. A representative intensity-based registration estimator solves the objective

$$\widehat{\boldsymbol{\varphi}}_{0\to t} = \arg\min_{\boldsymbol{\varphi}_{0\to t}} \mathcal{D}(I_0,\, I_t \circ \boldsymbol{\varphi}_{0\to t}) + \alpha\, \mathcal{R}(\boldsymbol{\varphi}_{0\to t}), \tag{3}$$

in which $\mathcal{D}$ rewards image agreement, $\mathcal{R}$ encodes prior structure and α sets their balance. The map is written here as a forward map on the reference domain. Frameworks that optimise a backward sampling map use a different convention, and the two conventions must not be interchanged when transformations are composed or differentiated for strain. Groupwise registration extends such an objective over a sequence, while an amortised registration network can minimise its expectation during training. Other families use different targets: StrainNet learns from DENSE-derived displacement, DLSS uses velocity and segmentation supervision, and a downstream disease model learns from clinical labels. Their training

losses, inference mappings and evaluation references must therefore be distinguished. No universal image-matching loss unites these methods. What they share is the need to combine observed information with representational and learned assumptions.

Where intramyocardial texture is weak, different fields may achieve nearly the same image agreement while implying different displacement gradients. The prior then helps select the estimated field; Figure 2 demonstrates the narrower case in which only masks constrain correspondence. Three factors require separate tests. Input information limits what can be identified from the images. Representation limits which fields the model can express. Estimation determines which admissible field the optimiser or network recovers. These factors can interact: a bandwidth restriction, for example, changes both representational capacity and optimisation, and displacement agreement does not guarantee agreement of its spatial derivatives.

A direct fit to a known field probes the representation under a stated norm, sampling scheme and boundary treatment, with displacement and the target strain or abnormality attribute evaluated separately. Running the estimator on rendered images probes recovery under the combined image and estimation conditions. Changing regularisation, resolution or views can expose sensitivity, but causal attribution needs matched controls and retraining when a training-loss weight changes. These experiments diagnose possible mechanisms; they do not assign errors to mutually exclusive causes. The interpretation of representation fitting is developed in Section 5.3, and the proposed programme specifies the additional optimisation checks needed before a failed fit can establish a capacity limit.

Temporal strategy adds a further assumption to the objective. Reference-to-frame tracking keeps one reference configuration and must therefore estimate large displacements from it; sequential tracking estimates smaller inter-frame steps and accumulates composition error and drift; groupwise methods estimate all phases jointly under whatever temporal constraint they impose [26, 27]. None of these strategies guarantees correct intermediate phases, because a periodic deformation can close perfectly at the end of the cycle and still be wrong in between. Cycle closure is therefore a useful constraint and a useful quality-control signal, and no more than that.

3.2 Feature tracking and contour propagation

Feature-tracking (FT) software is attractive because it operates on routine cine and fits existing clinical workflows. Initial contours are propagated through the cardiac cycle, after which strain may be calculated from endocardial length, a myocardial mesh or a vendor-specific deformation field; interior motion is to a large degree interpolated from boundary behaviour [13]. Because each of these steps can be implemented differently, the phrase “feature tracking” denotes a family of measurements. Commercial implementations have differed in contour placement, transmural treatment, radial-strain calculation, temporal interpolation and user correction. As a consequence, values may fail to be interchangeable across products or software versions even when repeatability within a product is acceptable [16, 28].

The family’s clinical validation began before deep learning. Maret and colleagues related cine-derived functional measurements to scar in 30 patients after ST-elevation infarction [29]; Hor and colleagues compared mid-ventricular whole-slice peak circumferential strain with HARP in 191 patients with Duchenne muscular dystrophy and 42 controls [30]; and Augustine and colleagues studied 145 healthy volunteers with a tagging comparison in 20, finding closer global circumferential than longitudinal or radial agreement [31]. These are abstract-level readings, and the three studies validated three different regional quantities. Deep learning entered the family through its front end. Vigneault and colleagues segmented the sequence with a fully convolutional network and fitted quadratic basis splines across all frames in 42 patients with hypertrophic cardiomyopathy and 21 controls [32], and Puyol-Antón and colleagues

combined convolutional segmentation with subsequent feature tracking and strain calculation [33]. In both designs the learned component is the segmentation, and the motion model remains a conventional tracker.

The strongest interpretation of a contour-tracking result depends on what was actually tracked. A reliable change in endocardial length can support a global or segment-level contour-derived strain under that definition; it cannot by itself support transmural material strain. The converse caution also applies: a method that uses internal image information should be described by its documented algorithm, and a product-category label such as "border tracking" should not replace that description. Method descriptions and validation endpoints, then, should govern interpretation. One evaluation caveat applies across Section 3. Segmentation can initialise the search region, define the cardiac coordinate system and, in several learning-based designs, supply an anatomical loss during training. When the same reference contours both shape the fitted transformation and provide the overlap score that assesses it, the evaluation is not independent of the fitting.

3.3 Conventional and learned registration, and the field representation

Intensity-based deformable registration estimates a dense field by optimising Equation (3) directly. The free-form deformation model of Rueckert and colleagues, an affine transform plus a B-spline control lattice with a smoothness penalty, is a widely used representation originating in breast MRI [34]. Groupwise variants remove the dependence on an arbitrarily chosen reference frame. Metz and colleagues fitted an nD+t B-spline model to the whole dynamic sequence with spatial and temporal smoothness and an optional cyclic constraint, reporting sub-voxel accuracy on synthetic and thoracic CT data [26]. More recently, Chen and colleagues replaced the groupwise dissimilarity term with a locally low-rank metric and evaluated it on a 20-case simulation, 100 public cine cases and 16 in-house tagging patients. They reported reduced late-diastolic drift and better correlation with tagging for radial strain in several segments than Farneback optical flow, pairwise registration and global low-rank groupwise registration. Comparisons with nD+t B-splines and pTVreg concerned simulation and in-vivo tracking endpoints [27]. Optical flow itself offers another route to local correspondence: Wang and colleagues combined intensity and shape-feature invariance in a gradient-based method for cine and tagged MR, evaluated on realistic simulations and preliminary clinical data [35]. Two further designs supply correspondence through geometry or mechanics without an image-similarity term: a Laplace-based mapping between successive contours [36] and variational assimilation of cine observations into a contraction model [37]. The Laplace method was tested on synthetic data with in-vivo feasibility. For variational assimilation, the original abstract reports known-motion synthetic tests and 47 patients imaged at basal, mid-ventricular and apical levels, radial and circumferential strain agreement with tag tracking (intraclass correlation coefficient, ICC, above 0.8), and associations with clinical wall-motion scores. These positive findings remain abstract-level evidence here; the inferred trajectories in both approaches depend on their modelling assumptions. Anatomy-aware, pyramidal and online-adaptive methods change the problem along the same lines, by altering the evidence available to the optimiser and the class of admissible fields [38–40].

The learning-based literature began by amortising Equation (3). VoxelMorph reformulated registration as a function, parameterised by a convolutional network, that maps an image pair to a deformation field; the unsupervised training objective is the same image-matching and smoothness objective that a classical method optimises per pair, and the gain is that inference becomes a single forward pass [41]. TransMorph replaced the convolutional encoder with a hybrid Transformer–convolutional architecture and added diffeomorphic and Bayesian variants, validated on brain MRI and phantom-to-CT registration [42]. In this family the prior is still smoothness or diffeomorphism, now partly learned, and the reported evidence is label overlap and deformation-field regularity.

A distinct strand places the constraint in the representation itself. B-splines, finite-element bases, Fourier coefficients, implicit neural representations and dense voxel grids have different capacity and conditioning. Fourier-Net predicts a band-limited representation of the displacement in a low-dimensional frequency domain and recovers the full-resolution field by zero-padding and an inverse discrete Fourier transform [43]; Fourier-Net+ takes band-limited representations of the images themselves as input, with cascaded and diffeomorphic variants [44]. Bandwidth explicitly constrains the base representation, making it a useful variable for the attribution experiment in Section 3.1. It does not necessarily constrain the final displacement to the same spectral support: Fourier-Net reports that scaling-and-squaring in its diffeomorphic variant produces a field that is no longer band-limited, and composition in cascaded variants also requires separate assessment. The implications for focal strain therefore depend on the complete mapping; the retained coefficients alone do not determine them. Both networks were developed as general registration methods. Fourier-Net+'s published abstract reports an evaluation on five datasets, including one 3D cardiac MRI dataset on which its diffeomorphic variant outperformed TransMorph and other baselines in Dice, Hausdorff distance, end-diastolic volume and ejection fraction. These are registration, geometric and functional measures; they do not by themselves establish regional material-strain fidelity. Direct evaluation of a band-limited cine estimator against a regional motion or strain reference is therefore a priority for the comparison proposed here, and the incomplete historical search record does not support any broader claim that such an evaluation is absent.

3.4 Cardiac-specific designs: joint tasks, multiple views, meshes and time

A second group builds cardiac structure into the estimator. Qin and colleagues learned motion estimation and segmentation jointly, sharing a multi-scale encoder between an unsupervised Siamese recurrent spatial-transformer branch and a segmentation branch, in a total cohort of 220 subjects, including 100 held-out test subjects [45]. Zheng and colleagues learned a pixel-wise "apparent flow" between two cine phases semi-supervised from a large unsegmented and a small expert-segmented set, then combined motion features derived from segmental radius and thickness with shape, volume and ejection-fraction features. The nine-feature classifier achieved 95% accuracy in five-fold cross-validation on the Automated Cardiac Diagnosis Challenge (ACDC) training cohort and 94% on the held-out challenge test set [46]. In that study the motion representation is validated through its usefulness for a downstream label, a pattern that recurs in Section 3.6. Motion Pyramid Networks (MPN) introduced multi-scale fused estimation with cyclic teacher–student training. In addition to mask comparisons, it measured endpoint and keypoint-tracking errors after applying motion generated by a separate network to images and initial keypoints; this is known imposed motion from a model; it does not supply an independent physiological reference [39]. FOAL added online adaptation of a learned estimator to a new sequence at test time [40]. A 2025 study used a memory-based model over all cine frames and evaluated supervision choices in 271 patients. Its abstract reported EPEs of 1.02 mm for the base model, 0.95 mm with end-diastolic distance-map supervision and 0.91 mm with full supervision, compared with 1.19 mm for a retrained RAFT baseline. Its evaluation points were manually initialised on the first-frame contours and propagated by conventional feature tracking; they were not motion-training targets. The reported EPE therefore measures agreement with those tracked points. Global peak-strain correlations provide a separate endpoint [47].

Multi-view and mesh-based designs target through-plane motion. MulViMotion fuses short-axis and long-axis cine in a hybrid 2D/3D network with a shape-regularisation module providing weak supervision, using 580 UK Biobank subjects split into 450 for training, 50 for validation and 80 for testing [48]. DeepMesh represents the heart as a 3D mesh of endocardial and epicardial surfaces, propagates a template mesh into subject space and estimates vertex-wise displacement through a differentiable mesh-to-image

rasteriser, maintaining temporal vertex correspondence and common template indexing across subjects; that indexing does not establish material homology [49]. CardioMorphNet takes the shape idea further: a recurrent variational autoencoder with separate posteriors for segmentation and motion, trained by recursively registering segmentation maps without an intensity-based similarity loss, validated on UK Biobank and M&Ms by comparing warped masks with reference masks and producing voxel-wise uncertainty maps [50]. Predictive uncertainty and accurate warped masks are valuable outputs, with two limits: the uncertainty is not automatically a calibrated regional error, and a Bayesian output is not automatically an abstention policy. STCMT-Net represents cardiac motion through keypoints extracted without supervision and reconstructs the dense field as a weighted combination of basis vectors from matched keypoint pairs, evaluated on 4D CT and cine MRI with a strain analysis reported to separate patients under different conditions [51].

Temporal structure has been addressed in parallel. Bidirectional recurrent registration predicts reference-to-frame motion with sequence context [52]; SegMorph couples motion and segmentation in a recurrent variational representation [53]; and IG-BiReg addresses implicit-guided bidirectional residual registration, but its detailed mechanism and performance could not be verified from an accessible primary source [54]. A further variant is a Mamba-based sequential model presented in a MICCAI workshop [55]. A continuous-time latent neural ordinary differential equation adds bidirectional forward–backward evolution and a Lagrangian consistency term, evaluated on ACDC and M&Ms [56]. One parallel development belongs here because of what it leaves out. CMR-specific foundation models now exist, one pre-trained on 15 million cine images from 74,916 studies [57] and another trained by self-distillation on 36 million images from 27,524 subjects [58], each fine-tuned for segmentation, landmark localisation, diagnosis and related tasks. The retrieved task descriptions do not include dense motion estimation. Their segmentation and clinical-task results establish useful representations, while motion-specific transfer remains outside the evidence inspected here.

Together, these studies have made progress on three-dimensional motion from two-dimensional acquisitions and on temporal consistency across the cycle. Their reported validation needs stating plainly. In the cardiac-specific papers reviewed here, the dominant evidence is agreement between a warped segmentation and a reference segmentation, together with surface distances and deformation-field regularity, with landmark error where a landmarked dataset was available. That evidence is appropriate for the claims those papers make, and it is also the evidence that Figure 2 shows cannot by itself establish material correspondence.

3.5 Biomechanical and physics-informed regularisation

A third group replaces a generic smoothness penalty with a statement about tissue mechanics or about the function space itself. Qin and colleagues learned the prior from data: a variational autoencoder trained on biomechanically simulated deformations defines a manifold of plausible fields, and tracking becomes a search over that manifold, validated on three public cine datasets [59]. WarpPINN states the physics explicitly. It represents the deformation with a neural network fitted per case under an intensity-similarity term plus a hyperelastic regulariser and a Jacobian penalty for near-incompressibility. Strain follows by automatic differentiation, and Fourier feature mappings allow sharp features in the strain field. It was tested on synthetic examples and a cine benchmark of 15 healthy volunteers [60]. WarpPINN-fibers adds a fibre-stretch penalty and tests displacement recovery against prescribed motion in a synthetic phantom. On the same healthy benchmark it reports improved landmark tracking and compares estimated strain curves for physiological plausibility; no reference strain curves were available to establish improved human strain accuracy [61]. Liu and colleagues combine the mechanical and temporal arguments in a space–time

finite-element correlation framework with region-specific biomechanical regularisation and a data-driven temporal decomposition. The published abstract reports evaluation against two feature-tracking and four deep-learning baselines using one synthetic dataset with known motion and strain, three public datasets with landmarks or masks and one clinical dataset with masks. The known-motion tests concern 2D+t; 3D+t motion was evaluated only on the clinical dataset using masks [62].

Such results show that mechanics can be useful inductive structure. They do not imply that a constrained field is uniquely correct, because disease can alter wall thickness, loading, scar mechanics and fibre organisation, and a 2D slice omits through-plane motion. The studies summarised here combine controlled synthetic tests with in-vivo landmark or mask evaluation. These targets establish different properties, and the available summaries do not resolve focal strain recovery across pathological conditions. A mechanical prior is a differently shaped constraint from a smoothness prior, neither weaker nor stronger in general, and whether it preserves a focal deficit depends on whether its constitutive assumptions hold in heterogeneous injured tissue. That dependence is an open question, and validation must test it directly by asking whether the constraint preserves the abnormality of interest, in addition to assessing plausibility.

3.6 Learning from cine, contours or motion-encoded CMR

The following examples distinguish the acquisition used at inference, the learning target and the endpoint used for evaluation. They include cine pipelines and methods that analyse the material-sensitive comparator itself. DeepTag estimates motion on tagged images without labels, using bidirectional diffeomorphic registration between consecutive frames and a differentiable composition layer to recover the Lagrangian field, validated by landmark tracking on a clinical tagging dataset [63]. Networks trained on synthetic 3D tagged images from a biophysical left-ventricular model have since been evaluated for 3D tagging analysis, including an external human landmark comparison and porcine strain comparisons with manual cine annotations [64]; like DeepTag, they operate on the comparator acquisition and not on cine. DeepStrain provides an end-to-end cine workflow. It centres the ventricle, segments anatomy, estimates end-diastole-to-phase displacement and derives Green–Lagrange radial and circumferential strain, and the original study reported a Cardiac Motion Analysis Challenge (CMAC) landmark end-point error of 2.9 ± 1.5 mm together with scan–rescan repeatability in ten healthy volunteers and high agreement for volumetric measures [65]. A later single-centre comparison in 47 healthy participants and 533 patients found correlations with manual feature-tracking analysis of 0.83–0.85 for global radial and 0.88–0.91 for global circumferential strain [66]; in that comparison the comparator is another estimator. These results establish workflow and output properties. Independent dense material truth for the cine motion field is a separate matter, and a prescribed-motion evaluation of DeepStrain is discussed in Section 4.1.

StrainNet is particularly relevant because DENSE phase-derived displacement was the supervised target, and its two evaluations must be read separately. Of 305 eligible individuals (161 patients, 99 healthy adults and 45 healthy children or adolescents), 243 individuals contributing 670 sections formed the training set and 62 individuals contributing 190 sections formed the participant-held-out internal test set; the 305 were not independent held-out cine tests. For pixelwise displacement end-point error (EPE), the model was deliberately given contours derived from DENSE magnitude images, permitting one-to-one comparison with DENSE phase-derived displacement over the myocardium and time series; the held-out-test EPE was 0.75 ± 0.35 mm. In the distinct routine-cine analysis, contours from bSSFP cine feature tracking were supplied to StrainNet and registered DENSE was the comparator. End-systolic global and AHA-segmental circumferential Green–Lagrange strain was analysed in 59 of the 62 test participants with basal, mid-ventricular and apical images: ICCs were 0.87 versus 0.72 globally and 0.75 versus 0.48 segmentally for StrainNet versus conventional feature tracking [18]. The improvement in ICC

was accompanied by prose-reported cine–DENSE limits of agreement of −5.74 to 4.72 percentage points globally and −9.16 to 8.07 percentage points segmentally. A small discrepancy between the source prose and table for the global limits is documented in Supplementary Table S1. The study used an internal participant split, with cine-mask preprocessing and resolution matching; external-dataset validation was not performed. These results support improved end-systolic Ecc agreement, while the remaining individual differences require comparison with an intended-use tolerance. The two tests retain distinct interpretations: the EPE is not a cine-contour result, "global" strain is not the pixelwise whole-myocardium EPE, and end-systolic values are not peak-strain timing. The inference therefore remains model-, direction-, population-, acquisition- and registration-specific, and it does not establish focal location or extent.

The same supervision route has since been extended. Lei and colleagues describe a DENSE-supervised model with separate myocardial-rotation and radial-motion branches for routine bSSFP cine; the retrieved abstract motivates this design by difficulty recovering large rotation [67]. MyoNet takes a related approach in Dahl salt-sensitive rats with radiation-induced injury. Cine segmentations are converted to binary spatiotemporal inputs, and displacement supervision is obtained by applying sine-wave modelling (SinMod) to paired tagged MRI. Reported circumferential and radial strain ICCs were 0.973 and 0.975 [68]. This is evidence for a segmentation-to-motion learning route in a preclinical setting, with an analysis-derived tagging reference; it does not establish accuracy in human routine cine.

Clinical labels enter these pipelines at different stages. Cine-only infarct delineation combines recurrent motion features and optical flow with manual delayed-enhancement delineations as the target [69]. Deep-learning synthetic strain (DLSS) predicts myocardial velocities and segmentation, using velocity supervision from registered 4D-flow MRI and segmentation targets generated by commercial software, and derives strain from cine at inference. Its development dataset contained 223 examinations from 219 patients; four-reader consensus wall-motion labels were used for the separate clinical evaluation in 53 patients and 846 segments. The ROC AUC was 0.90; a 30% peak-radial-strain threshold gave sensitivity 86% and specificity 85% [23]. Thus the reader labels evaluate the clinical output; they do not supervise the motion network.

Deep-learning aligned strain also separates motion estimation from clinical classification. Its cine pipeline detects key phases, segments the myocardium and estimates deformation using image similarity, smoothness and mask agreement; late gadolinium enhancement (LGE) labels the downstream patient-level fibrosis classifier, and the displacement field is learned without LGE. Development used 57 patients with Duchenne muscular dystrophy, followed by a same-centre temporal holdout of 82 patients. The classification target was the presence of at least one LGE-positive segment; the spatial boundary of fibrosis was not a target [70]. In a different chamber and population, Bello and colleagues used right-ventricular motion to predict survival in 302 patients with chronic thromboembolic pulmonary hypertension. Internally bootstrap-corrected Harrell C indices were 0.75 for the learned model and 0.59 for a model using manually measured conventional volumetric and functional variables [22]. Both studies provide clinically relevant discrimination evidence in their tested settings, without converting that evidence into dense motion accuracy or demonstrated treatment benefit.

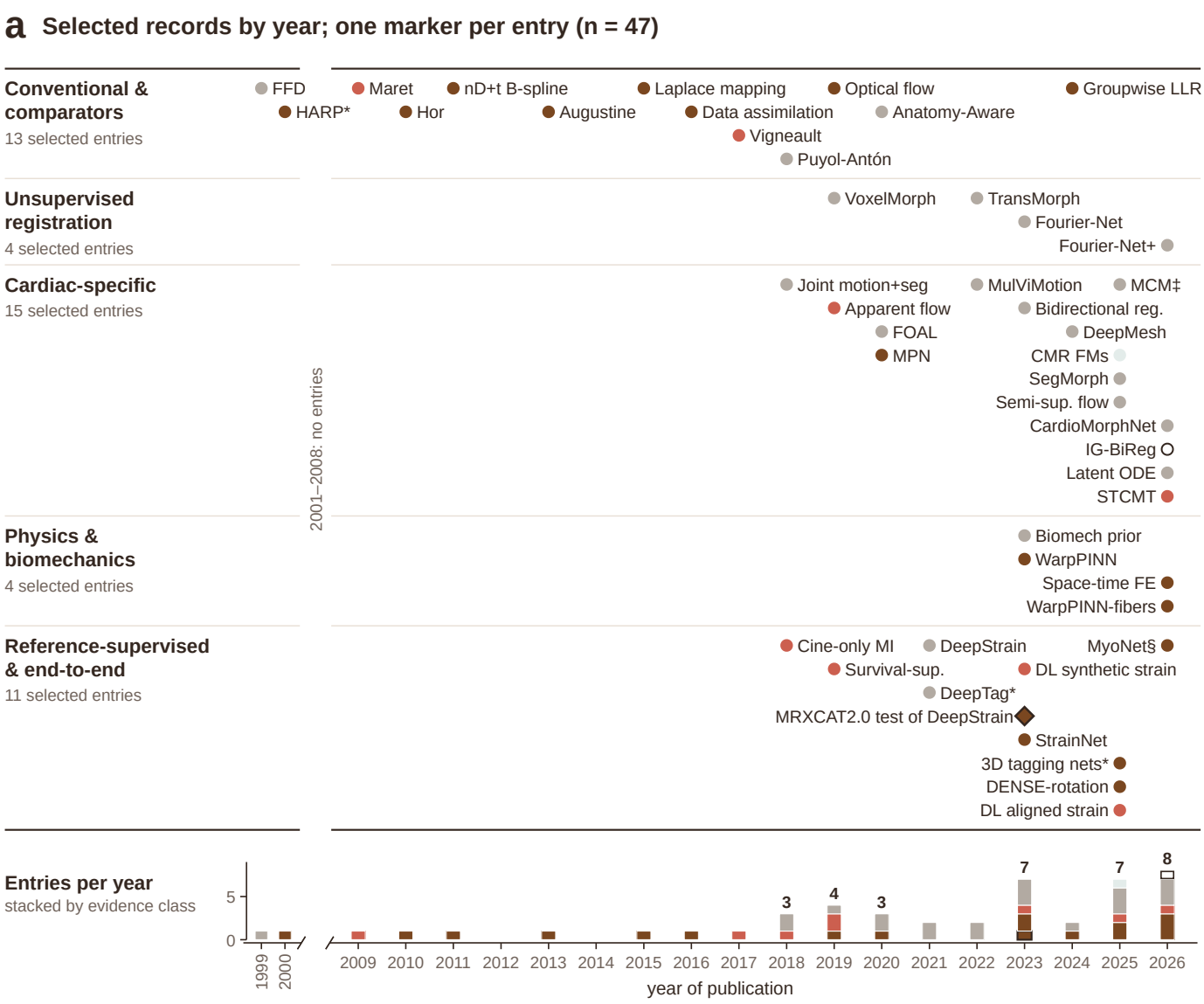


Figure 3. Selected method and evaluation records by year (Supplementary Table S3). One marker denotes a plotted entry; entries are neither independent cohorts nor unique algorithms. Related non-cine methods and separate evaluations are included. The three early clinical feature-tracking studies and the MRXCAT2.0 evaluation of DeepStrain are plotted as separate entries, and the generic feature-tracking family row of Table S3 is not plotted. Colours identify one selected documented endpoint per entry; studies may report several endpoints. They do not rank validation strength. Motion Pyramid Networks (MPN) include model-generated imposed-motion tests, which are distinct from independent physiological references. "Endpoint not verified" and "other-task validation" are separate categories, neither of which means that a study performed no validation. Source-specific reading depths and populations are documented in the figure data. The selective map is not a prevalence estimate for the field. Years 2001–2008, which have no selected entries, are compressed.

4 What the validation literature establishes

No single study can answer every layer of the measurement chain, so the most informative synthesis groups evidence by validation target; algorithm chronology, the organising axis of Section 3, is set aside here. Figure 5a summarises what each target can establish. The following subsections take the targets in turn, from the most controlled to the most clinical.

4.1 Known-motion tests establish capacity and technical error under controlled conditions

Numerical phantoms make exact correspondence available. A simple realistic phantom showed that spatial and temporal resolution can materially affect feature-tracking measurements [71]. MRXCAT2.0 couples statistical shape, biophysical simulation and tissue texture to produce healthy and pathological cine-like images paired with known functional ground truth [15]. Other in-silico heart models and DENSE-SIM extend controlled testing of strain pipelines and acquisition effects [72, 73]. The design that such resources make possible, a graded regional deficit measured against a material reference while the estimator is held fixed, has a precedent outside cine: Yeon and colleagues compared tagged circumferential shortening with implanted sonomicrometry crystals in a canine coronary-ligation model, with both methods separating ischaemic from remote myocardium [74]. This provides a precedent for regional reference-based testing, although it is not a controlled sweep of lesion size or severity in routine cine.

MRXCAT2.0 provides a useful prescribed-focal example: an evaluation of DeepStrain by a separate development team. Four left-ventricular cases were generated (normal, dilated, hypertrophic and infarcted; ejection fractions 51%, 34%, 41% and 49%). An elliptical free-wall scar was prescribed in the infarct case, while remote compensation preserved its ejection fraction. The underlying abnormality differed by component: remote-to-scar peak systolic strain changed from 0.95 to 0.30 radially (Err), from -0.18 to 0.01 circumferentially (Ecc) and from -0.18 to -0.17 longitudinally (Ell). Radial strain was markedly reduced but retained a substantial residual value; circumferential strain approached zero, while longitudinal strain changed little. These are prescribed tissue values; the estimator's errors are reported separately below [15].

DeepStrain achieved a reported myocardial Dice of 0.82 across phases and displacement error of 1.0 ± 0.9 mm. The reported circumferential strain error was 0.02 ± 0.04, compared with radial strain error of -0.24 ± 0.21 across cases and -0.20 ± 0.21 in the infarct case. The source's Figure 14 averages the measures over three mid-ventricular short-axis slices; these summaries do not quantify error within the scar. Although the source describes the infarct case as the lowest performer, with its lowest Dice in the thin scar region, the infarct-case radial mean error is smaller in magnitude than the all-case mean. Neither these means nor overlapping standard deviations establish a uniform case ranking or equivalence. Different metrics can rank cases differently, and the all-case summary includes the infarct case. Figure 4 preserves the reported values while separating them from unmeasured scar attributes.

The generated domain also limits interpretation. The evaluation used one scar geometry and a signal-to-noise ratio (SNR) of 30, and myocardial texture was removed during image generation. The MRXCAT2.0 texture network (TextNet) and DeepStrain were both developed with ACDC data: independence of the evaluating team does not establish independence of the image domain, although shared dataset use alone is not evidence of leakage. The inspected source does not identify the evaluated DeepStrain release or weights. The component-specific errors could reflect image formation, wall thinning and partial volume, training data or estimation constraints; the experiment does not isolate these contributions [15].

Known-motion resources allow investigators to test representation capacity, recovery at a chosen signal-to-noise ratio and sensitivity to sampling or regularisation. Existing work has varied resolution [71], varied analysis regularisation [73], or represented regional pathology in simulated motion [15, 75]. These

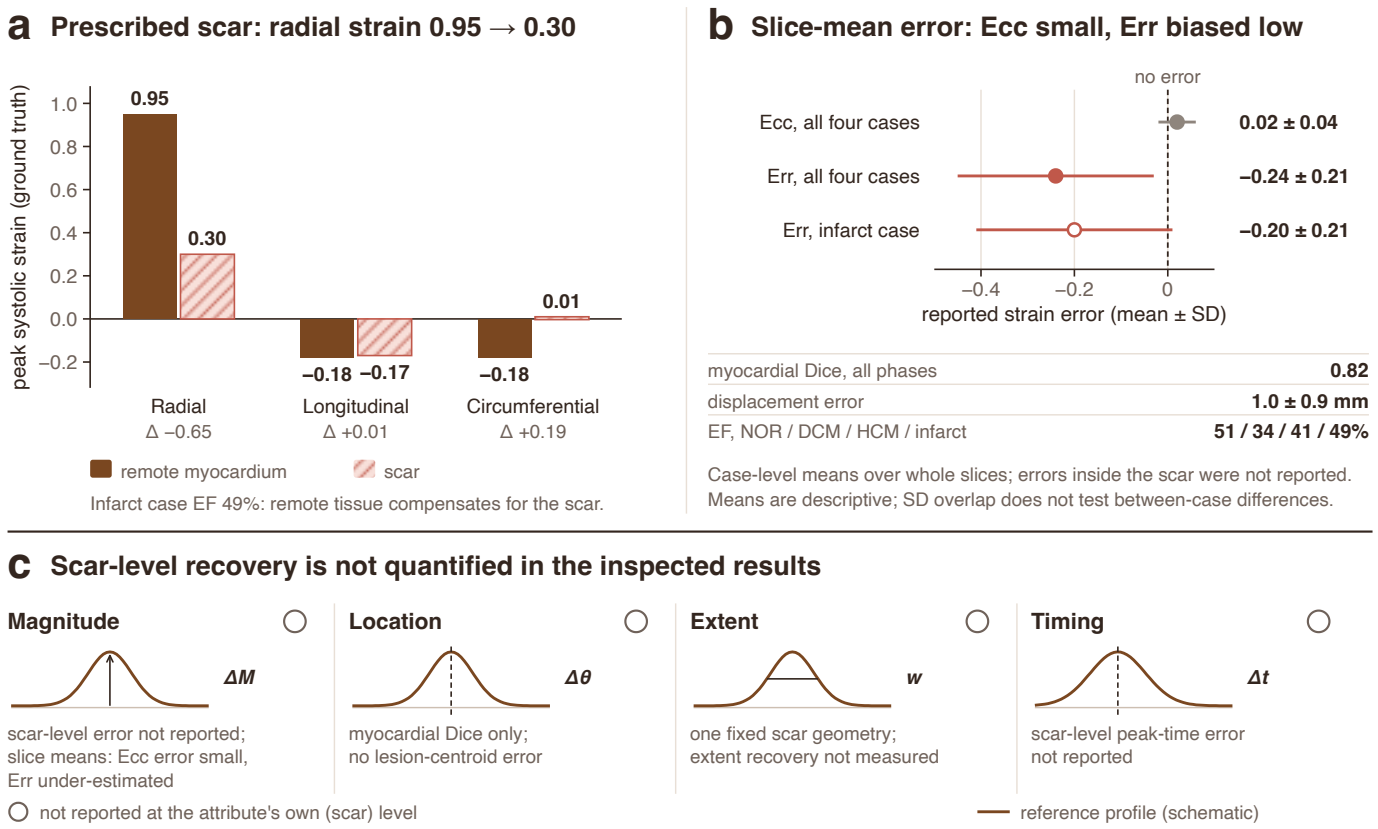


Figure 4. Prescribed tissue strain and reported DeepStrain errors in MRXCAT2.0. (a) Remote-to-scar peak systolic strain: radial strain is markedly reduced (0.95 to 0.30), circumferential strain approaches zero (−0.18 to 0.01), and longitudinal strain changes little (−0.18 to −0.17); infarct-case ejection fraction (EF) is 49%. Differences are arithmetic scar-minus-remote values. (b) Source-reported means and standard deviations (SD): Ecc error 0.02 ± 0.04, Err error −0.24 ± 0.21 across cases and −0.20 ± 0.21 in the infarct case. Source Figure 14 averages over three mid-ventricular short-axis slices. The all-case summary includes the infarct case; SD overlap does not establish equivalence, and no uniform case ranking is inferred. (c) The inspected results do not quantify scar-level magnitude, localisation, extent or peak-time error. Profile glyphs define attributes and are conceptual. Source values are retained [15]; the redraw adds no new experiment. The separate evaluating team and shared ACDC image domain are discussed in the text.

are useful precedents for the proposed validation programme. The surviving search records do not establish whether all combinations of graded lesion size, severity, transmurality and fixed-estimator evaluation have already been studied. Here, graded attribute recovery is proposed as a research priority; the proposal does not rest on a claim of field-wide absence. Simulation results remain conditional on the anatomy, mechanics, image formation and pathology represented; paired in-vivo comparisons are needed to assess transfer.

4.2 Paired tagging or DENSE provides material-sensitive in-vivo evidence

Tagging introduces a prepared magnetisation pattern; DENSE encodes displacement in image phase. Both make myocardial motion more directly observable than routine cine, and neither is error-free. Tag fading, spatial filtering, phase unwrapping, signal-to-noise, through-plane motion, temporal fitting and registration to cine can all contribute reference error. A comparison should therefore report reference uncertainty and acquisition alignment, and it should avoid calling either technique literal ground truth.

The paired literature is mixed in a scientifically useful way. In three healthy volunteers and 41 patients with recent or chronic myocardial infarction, Gao and colleagues reported agreement in peak circumferential strain and similar regional patterns between cine registration and DENSE, while cine radial strain was overestimated [76]. A larger comparison of 77 patients with cardiomyopathy and ten controls found relatively small circumferential differences but wide radial differences across tagging, DENSE and three feature-tracking products [17]. Wehner and colleagues showed that part of the apparent cine–DENSE discrepancy can arise from comparing different strain definitions, while still finding that regional tracking information cannot be inferred from contours alone [25].

The product dependence seen by Cao and colleagues was examined directly at 3 T by Militaru and colleagues, who compared three feature-tracking products with tagging in 61 participants. Global longitudinal and circumferential agreement was high, regional and radial agreement varied more, and scar discrimination differed by vendor [16]. This pattern supports global and some regional use under specified products, and it warns against interchangeability and against extrapolating global performance to radial or highly local outputs; it does not justify a blanket rejection of feature tracking. Among the paired studies reviewed here, timing was compared less often than magnitude. In 42 participants, Brandt and colleagues found moderate-to-good agreement for global and segmental circumferential time to peak strain between tagging and feature tracking, alongside significant biases in segmental peak strain [77]. This supports a segment-level timing comparison; a focal peak-time error endpoint was not established by the abstract inspected here.

A 2026 study makes the importance of the endpoint explicit. Jahromi and colleagues compared one- and two-slice DENSE models with a one-slice cine model in a deforming phantom and 40 healthy volunteers. In the volunteer comparison, the endpoint was slice-wise peak median systolic endocardial circumferential strain at two mid-ventricular levels. The cine versus two-slice DENSE mean difference was -2.7 percentage points. Two one-sided tests supported equivalence within a selected margin of ± 3.5 percentage points, despite a significant difference from zero; those are compatible findings because they test different hypotheses. Prospective specification of the margin and the handling of within-person slice correlation were not established by the inspected report. The radial, longitudinal and epicardial circumferential comparisons differed [78]. Thus the result supports a specific healthy-volunteer, slice-level, endocardial comparison; it says nothing about focal-lesion accuracy or general interchangeability. Selected studies are compared in Table 2.

Table 2. Empirical validation studies relevant to regional fidelity. The analysis unit and support boundary are retained because participant, slice, segment and pixel results are not interchangeable.

Study; development/test denominator	Population/input; statistical unit	Spatial/temporal estimand	Comparator	Result, including uncertainty where reported	Supported scope
Wehner et al. [25]; 88 subjects, 186 paired slices	Paired cine/DENSE at two centres; subject/slice	Whole-slice base/mid/apex Ecc; longitudinal strain, torsion and dyssynchrony	DENSE; contour-only calculation	Mid-slice Ecc bias -0.4 percentage points (absolute) and coefficient of variation 11%; longitudinal strain bias -0.2 percentage points, coefficient of variation 19%; dyssynchrony bias 26 ms, coefficient of variation 76%	Selected whole-slice quantities, not focal localisation or extent
Cao et al. [17]; 87 subjects	77 cardiomyopathy, 10 controls; subject; matched mid-short-axis and four-chamber planes	Peak Ecc, Err and Ell from DENSE and three FT products	Tagging	DENSE Ecc agreement with tagging ICC 0.778; technique/product differences were much wider for Err than Ecc/Ell	Peak Ecc/Err in one mid-short-axis slice and Ell in four-chamber; no dense location/extent error
Militaru et al. [16]; 61 subjects	18 healthy, 18 chronic myocardial infarction, 15 dilated cardiomyopathy, 10 LV hypertrophy; subject/segment	Global and 16-segment transmural peak systolic LS/CS/RS	Tagging; LGE in MI	Versus tagging, global ICC 0.92–0.94 for LS and 0.88–0.91 for CS; global RS ICC 0.10–0.63 versus tagging and 0.10–0.81 between FT products; regional differences were larger	Strong global LS/CS under tested products; weaker regional/radial interchangeability
Kihlberg et al. [21]; 116 patients	Suspected coronary artery disease; subject/segment	ROC endpoint: end-systolic segmental Ecc	LGE $>$ 50% transmurality	34 patients had qualifying scar; segment AUC 0.87 DENSE, 0.83 tagging, 0.66 FT; sensitivity at 80% specificity 82%, 71%, 35%	Scar discrimination at the stated threshold, not material-motion error
StrainNet [18]; 243/62 subjects (670/190 sections)	Patients; healthy adults and children; pixel, subject and AHA segment	DENSE-contour displacement over time; separately cine-contour end-systolic Green–Lagrange Ecc	DENSE phase displacement; conventional FT	Test EPE 0.75 ± 0.35 mm with DENSE contours; 59 cine tests: ICC 0.87 vs 0.72 global and 0.75 vs 0.48 segmental; cine–DENSE limits of agreement -5.74 to 4.72 and -9.16 to 8.07 percentage points	Bounded Ecc support; EPE is not ordinary-cine displacement error
Strain-8 [20]; 20 volunteers	Healthy same-day scan–rescan; subject/segment	Global and segmental peak strain across eight protocols	Repeat acquisition	Global coefficients of variation ranged from excellent to fair ($\leq$ 20%); segmental values were frequently poor ($>$ 20%), especially longitudinal/radial	Same-day precision in healthy volunteers, not accuracy

Study; development/test denominator	Population/input; statistical unit	Spatial/temporal estimand	Comparator	Result, including uncertainty where reported	Supported scope
MRXCAT 2.0 with DeepStrain [15, 65]	Generated normal, dilated, hypertrophic and infarcted cases (EF 51/34/41/49%); mid-ventricular short-axis slices	Known simulated peak radial, circumferential and longitudinal strain; pixel/mesh	Biophysical prescription with an elliptical free-wall scar	Dice 0.82; displacement error 1.0 ± 0.9 mm; mean Ecc error 0.02 ± 0.04 and mean Err error -0.24 ± 0.21 averaged across cases and slices (-0.20 ± 0.21 for the infarct case)	Component-specific capacity/error within the generated domain; slice-averaged, with no scar-region error reported; one scar geometry and one estimator; release not identified
Jahromi et al. [78]; phantom and 40 volunteers	Healthy, mid-ventricular; subject/layer/slice	Slice-wise Ecc, Ell and Err over normalised cycle	Deforming phantom; one-/two-slice DENSE models	Peak median systolic endocardial Ecc: mean difference -2.7 percentage points; two one-sided tests within selected ±3.5-point margin; other components/layers differed	Component- and layer-specific equivalence, not pathological dense motion

The studies in Table 2 do not all inspect the four attributes in the motivating question, so Table 3 records the direct endpoint of each, keeping a segmental analysis distinct from a localisation or extent validation.

Table 3. Direct inspection of abnormality magnitude, location, extent and timing in selected paired or regional studies. "Not established here" means that this review has not verified a direct endpoint for that attribute at its available reading depth. It does not assert that the original study omitted the analysis. Explicitly described tasks and full-text findings are distinguished in the cells.

Study	Magnitude	Location	Extent	Timing
Gao et al. [76]	Peak regional circumferential/radial strain; circumferential agreement, radial overestimation	Similar regional patterns reported at abstract depth	Lesion-boundary or width error not established here	Peak values; timing error not established here
Wehner et al. [25]	Base/mid/apex whole-slice Ecc, longitudinal strain and torsion; only mid-slice Ecc was judged acceptable by the study (coefficient of variation 11%, bias not significant)	Slice level, not focal-lesion localisation	Focal spatial extent not established here	Dyssynchrony was direct and showed poor cine–DENSE agreement
Militaru et al. [16]	Global and segmental peak strain; direction/product-dependent agreement	Segment-level scar discrimination was tested and varied by product	No lesion-boundary or lesion-size error	Peak endpoints; timing error not established here
Kihlberg et al. [21]	Global/segmental strain and torsion associated with LGE category	Segment-level discrimination of high-transmurality LGE was direct	A > 50% transmurality category is not a boundary/area error	Timing error not established here

Study	Magnitude	Location	Extent	Timing
StrainNet [18]	End-systolic global and segmental Ecc showed positive DENSE agreement	AHA segment averages; no focal localisation-error endpoint	Focal extent not established here	Fixed end-systolic comparison; peak-time error not established here
DeepStrain on MRXCAT2.0 [15]	Prescribed scar: case-averaged Ecc error 0.02 ± 0.04 versus Err error -0.24 ± 0.21; a prescribed focal example reported as slice averages, without a scar-region error in the inspected results	Lowest Dice reported in the thin scar region; no centroid or boundary-distance endpoint	Scar extent fixed by design; extent error not reported	Peak values; timing error not established here
Jahromi et al. [78]	Component-, layer- and slice-specific strain in phantom/healthy volunteers	No pathological lesion-localisation task	Focal extent not established here	Focal timing error not a reported endpoint

4.3 Repeatability establishes precision, and leaves accuracy open

Scan–rescan studies are essential for longitudinal use, because they show how much a result may change when the underlying subject is assumed stable. The Strain-8 study scanned 20 healthy volunteers twice on the same day using eight protocols: cine, tagging and DENSE at 1.5 and 3 T, plus 2D and 3D echocardiography. Global repeatability ranged from fair to excellent depending on direction, protocol and field strength, while segmental results were more variable [20]. A multicentre travelling-volunteer study found the same kind of variation across scanners and post-processing [79]. Two cautions follow from such studies. A repeatable biased method can still be inaccurate, and a method with good global repeatability can have weaker segmental precision. Both absolute error and variance should therefore be reported at the claimed scale.

4.4 Tissue references test biological correspondence, and leave motion truth open

LGE is an independently measured tissue characteristic with strong relevance to infarction, and it is not a displacement measurement. If regional strain differs between scarred and non-scarred segments, the result supports tissue association and possibly diagnostic discrimination. It leaves the pointwise displacement error unknown, because scar transmurality and deformation are related without being identical, and remote loading, tethering and compensation can alter strain.

Kihlberg and colleagues provide a clear example. In 116 patients with a high likelihood of coronary artery disease, 34 had more than 50% transmural LGE in at least one segment. For discriminating those segments, reported areas under the receiver operating characteristic curve (AUCs) were 0.87 for DENSE, 0.83 for tagging and 0.66 for feature tracking; at 80% specificity, sensitivities were 82%, 71% and 35%, respectively [21]. This is clinically relevant comparative evidence that the techniques captured scar-associated dysfunction differently in that cohort. Read as a direct ranking of material-motion accuracy, it would say more than it measured.

Buss and colleagues associated global feature-tracking strain with patient-level recovery of left-ventricular ejection fraction at six months after infarction [3]. He and colleagues evaluated segmental-strain measures for prognosis after ST-elevation infarction [80]. Such findings strengthen biological and clinical qualification. They also illustrate why motion accuracy and clinical usefulness should both be studied: a regionally smoothed estimate might retain prognostic signal, whereas a physically accurate field may fail to improve a decision if the endpoint or treatment pathway is poorly specified.

4.5 Outcome association is downstream evidence

Global or regional strain has been associated with adverse events across disease cohorts [5, 6]. Such associations establish prognostic value under the model, covariates and follow-up that were studied. Whether changing management on the basis of the measurement improves outcomes is a further question, and answering it requires a decision context, a comparator, a threshold, the consequences of error and preferably an impact evaluation. The distinction matters most for a regional map, where multiple testing, correlated segments and post hoc threshold selection can produce apparently focal findings.

5 Reconciling apparently contradictory findings

The literature looks contradictory when all endpoints are treated as interchangeable. Once differences in input, estimand and target are made explicit, the same literature becomes more coherent, and the following four subsections examine the main sources of apparent contradiction.

5.1 Global agreement and regional disagreement can coexist

Spatial averaging suppresses zero-mean local errors. If one segment is overestimated and another underestimated, a global mean may remain accurate. Boundary-constrained deformation can also reproduce total shortening while distributing it differently around the wall. For both reasons, strong global correlation and small global bias do not mathematically imply correct location or extent of a regional abnormality. Regional validation must therefore evaluate regional errors directly and account for correlated repeated segments within participants.

The reverse is also possible. A method may preserve the location of a focal reduction while showing a scale bias in absolute strain. Whether that output is useful depends on the intended task: localisation, longitudinal change, diagnostic threshold or mechanistic interpretation. A single correlation coefficient conceals these possibilities. Studies should report bias, limits of agreement and participant-level distributions for quantitative strain; localisation, extent and timing errors are additionally needed when those attributes form part of the intended claim.

5.2 Radial strain is a special stress test

Radial strain is repeatedly less consistent than circumferential or longitudinal strain [16, 17]. Three features of the measurement explain much of this. The ventricular wall spans relatively few voxels radially, trabeculation complicates the endocardial boundary, and a small contour displacement can produce a large relative change in thickness. Products may additionally calculate radial output from different entities. These factors make the definition and uncertainty of radial strain unusually important, without making radial function unknowable. Claims should therefore specify whether the result is tensor radial strain, wall thickening, a layer-to-layer distance change or another surrogate.

5.3 Regularisation can help and can bias

Smoothness, temporal bases, incompressibility and learned priors stabilise an ill-conditioned estimate. They can reduce noise and prevent folding, and they may also spread or attenuate a focal abnormality if the imposed scale is too coarse. Which of these effects dominates is an empirical matter: a smooth representation with sufficient basis functions can represent a focal defect, and an unconstrained dense field can still miss it because image cues are weak or optimisation fails. Three questions should therefore be separated:

1. **Representation capacity:** can any parameter setting express the target motion at the required scale?
2. **Estimation:** given images that contain sufficient cues, does the optimiser or trained model recover

those parameters?

3. **Input sufficiency:** do routine cine images contain enough information, under the tested acquisition and pathology, to distinguish the relevant motions?

Each question needs its own test. A representation-capacity test should not be replaced by a registration score, and failure on clinical cine should not automatically be attributed to the basis. In the other direction, success on noiseless synthetic images does not establish that the clinical input contains discriminating information. Section 3.1 describes controlled comparisons that probe these potentially coupled contributions, and the DeepStrain evaluation on MRXCAT2.0 in Section 4.1 shows why it is needed: the same estimator on the same images preserved one strain component better than another at the slice-average level. Nothing in that single experiment says which of the three was responsible.

5.4 Uncertainty and quality control answer a different question

An uncertainty map can describe posterior dispersion, ensemble variability or sensitivity to perturbation. It becomes a clinically interpretable error estimate only after calibration against relevant reference errors. Quality control stands in the same relation to accuracy: it can detect image artefact, contour failure or field folding, and it cannot guarantee that an accepted case has accurate regional strain. Methods should therefore report the observable signal used by a quality rule, its coverage, and error among accepted and rejected cases. A selective or abstaining workflow is appropriate only when such a rule is actually implemented, and it should not be imposed as a rhetorical requirement on every estimator. The consolidated mapping from validation target to supported claim is provided in Supplementary Note 2.

6 A proposed validation programme for regional or focal claims

A proposed evaluation programme should combine controlled tests, material-sensitive comparisons and clinical endpoints around an explicit measurement contract. Its objective is to match the strength of the claim to the evidence provided, which does not require every possible experiment from every paper. Figure 5b shows the resulting evaluation flow, and the subsections below describe its stages.

6.1 Pre-specify the measurement contract

Before comparing methods, investigators should state the cine protocol, views, temporal and spatial resolution, reconstruction, image-quality exclusions, abnormality type and scale, estimator version, user interaction, strain definition, reference alignment, target attributes and tolerances. Case splits should occur at participant and, where relevant, centre level. Repeated segments should not be analysed as independent observations. Hyperparameters, model selection and any quality threshold should be chosen without using the final test set. Competing methods should be tuned on the same data with comparable effort, because a retrained baseline and a published score answer different questions; and releasing the transformation direction, differentiation code and strain definition makes a disagreement diagnosable where a single aggregate score does not.

The primary endpoint should be an error in the claimed attribute, even when a surrogate is easier to obtain. If the claim is detection of a focal region, suitable endpoints include centre-of-abnormality distance, overlap or surface distance of the abnormal region, with sensitivity across lesion sizes. If the claim is quantitative severity, bias and limits of agreement for regional magnitude are required. If timing matters, peak-time or curve-alignment error should be assessed in addition to peak amplitude.

6.2 Stage 1: challenge representation and sampling with known motion

Known-motion experiments should vary one factor at a time and then test interactions: abnormality width, transmural extent, amplitude, onset/peak timing, position, wall thickness, through-plane motion, temporal resolution, spatial resolution, contrast, noise and artefact. The known field should first be fitted in the candidate representation under an explicit norm, sampling scheme and boundary treatment. The fitting objective must match the target: minimising displacement error need not minimise strain-gradient or focal-attribute error. A failed numerical fit alone does not establish a representation limit. Such a conclusion requires an analytic restriction, a certified approximation bound, or convincing evidence that the best attainable fit was reached; otherwise optimisation remains an alternative explanation. Both displacement error and error in the chosen strain or abnormality attribute should be reported. Running the estimator on rendered images then measures recovery under the combined image and estimation conditions; the difference between these tests is informative but is not automatically an additive or uniquely attributable optimisation error.

An equal-mean counterexample is particularly useful at this stage (Figure 5c). Two fields can have the same global strain while one contains a focal abnormality. Recovery of their mean is then no evidence of success; the method must distinguish their location and extent within pre-specified tolerances. Such tests show whether apparent global success masks spatial redistribution.

6.3 Stage 2: use paired material-sensitive in-vivo acquisitions

The material-sensitive comparison should pair routine cine with tagging or DENSE in the same session and align acquisitions carefully. Reference processing should be blinded to cine outputs where feasible. Registration between acquisitions, slice mismatch and temporal alignment should be quantified. Analyses should report results separately for healthy myocardium and relevant disease patterns, including scar location, wall thickness and image quality. External centres and scanners test transfer only if participants and acquisition conditions are genuinely independent of training and model selection.

No material-sensitive reference is perfect, so this stage also needs a sensitivity analysis. Varying reference smoothing, registration and segmentation estimates how much apparent cine error is attributable to the comparator. Where tagging and DENSE disagree, the disagreement should be reported; choosing the more favourable reference would hide it.

6.4 Stage 3: connect motion fidelity to tissue and decisions

Tissue characterisation, expert wall-motion scores and outcomes become valuable after the technical claim is clear. LGE can test whether a regional measurement captures scar-associated dysfunction; perfusion or mapping can address other substrates. Longitudinal studies can test responsiveness and minimal detectable change. Decision studies should specify what action changes, how false-positive and false-negative regional calls matter, and whether the regional measurement adds information beyond volumes, LGE and clinical covariates.

6.5 Validation resources answer different parts of the question

The resources reviewed here serve complementary purposes, and none supplies an error-free dense in-vivo reference across broad pathology. STRAUS supplies controlled multimodal images and framewise reference meshes; the Stanford repository supplies human acquisitions with closely matched cine, DENSE and tagging locations; CMAC supplies a historical cross-method benchmark with sparse material-sensitive landmarks; and MRXCAT generations supply programmable motion and acquisition conditions. For each, provenance and the unit of independence matter as much as file availability (Table 4).

STRAUS contains 18 *virtual patients*. These arise from three volunteer-derived multimodal templates,

a Validation targets support different claims

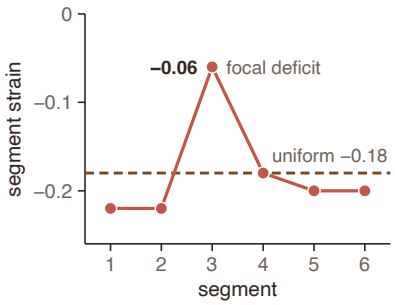
● Validation target	✓ Direct claim it supports	× Does not establish on its own
Known motion phantom, STRAUS, MRXCAT2.0	technical error in the simulated regime; capacity needs direct field fitting	in-vivo accuracy
Paired DENSE or tagging same session, registered	material-sensitive agreement under matched definitions	focal location or extent, unless measured
Scan–rescan same subject, re-imaged	precision; accuracy needs a reference	bias
LGE or tissue characterisation scar or fibrosis label	biological association; motion error stays unmeasured	pointwise displacement error
Outcome events, prognosis	prognostic association; utility needs a decision study	decision impact

b Proposed programme: combine complementary tests

Stage	What is varied or fixed	Resources (Table 4)
Pre-specify measurement contract	estimand, reference, tolerance and data splits fixed in advance	
Stage 1 Known motion	vary width, extent, amplitude, timing, resolution and noise	STRAUS **3 templates × 6 states** MRXCAT2.0 **programmable generator** CMAC **dynamic phantom**
Stage 2 Paired material reference	cine with DENSE or tagging in one session; registered; reference blinded	Stanford **55; pairs unverified** CMAC **15 volunteers, 12 landmarks**
Stage 3 Tissue and decisions	LGE, responsiveness, decision impact	Clinical cohorts **LGE or outcomes**

Counts are resource units, not independent test participants.

c Same mean, focal deficit



segments 1–3, focal	**−0.22 · −0.22 · −0.06**
segments 4–6, focal	**−0.18 · −0.20 · −0.20**
global mean, both fields	**−0.18**
range, focal / uniform	**0.16 / 0**

Recovering the mean does not establish deficit recovery.

Figure 5. What each validation target can establish, and a staged programme for regional or focal claims. (a) Five targets, a direct claim each supports, and what each leaves open on its own. (b) A proposed programme of complementary tests, with the resources reviewed in Table 4 placed alongside tests they can serve; counts denote resource units and are not counts of independent test participants. (c) Two fields with an equal global strain of −0.18 and different regional content (segment values −0.22, −0.22, −0.06, −0.18, −0.20 and −0.20 against a uniform −0.18): recovering the mean is no evidence of regional recovery. Conceptual synthesis; no study data.

each driven through one healthy and five pathological electromechanical states (left bundle branch block, left-anterior-descending, left-circumflex and two right-coronary-artery ischaemia patterns), so the 18 cases share three anatomies. Each virtual case provides 3D cine MR, three tagged-MR orientations, ultrasound and framewise meshes; the paper used engineering strain for its regional curves [75, 81]. The meshes therefore support known-correspondence displacement and strain calculations when coordinate system, reference frame, mesh identity and temporal sampling are matched. What they cannot supply is the full distribution of human anatomy, disease, image artefact or model error.

The Stanford Cardiac Performance MRI resource has a complementary strength, human acquisitions. Its laboratory page describes 51 healthy volunteers scanned at 3 T from 2023 to 2025, with whole-LV bSSFP cine and DENSE, tagging and mapping at specified corresponding locations [82]. DataCite metadata updated on 27 July 2026 describes 55 participants (the same 51 healthy volunteers plus four people with aortic stenosis) under DOI 10.25740/xt223hz2438. The two counts are versioned descriptions of one repository. Neither total proves that every sequence is complete and usable for every participant, that all cine–DENSE slices are registered analysis pairs, or that a derived paper has an independent test cohort. The DENSE magnitude and phase DICOMs also require reconstruction, phase unwrapping, displacement modelling, segmentation, spatial/temporal alignment and an explicit strain definition before they become a reference, so they are material-sensitive observations that become labels only after that processing. Jahromi and colleagues explicitly identify their healthy-volunteer data as part of this repository; individual subject-file correspondence was not verified. No subject files were downloaded or processed for this review.

CMAC paired cine SSFP, 3D tagged MRI and 3D ultrasound from a dynamic phantom and 15 healthy volunteers. Its human reference comprised 12 landmarks (four walls at three LV levels) manually tracked by two observers in 3D tagging, with a reported median inter-observer variability of 0.84 mm [83, 84]. That reference supports correspondence error at those landmarks and times, and it says nothing about dense strain accuracy between them. The MRXCAT family differs again. Original MRXCAT uses the XCAT phantom to generate cine or perfusion CMR with selectable cardiac/respiratory motion, sampling, coils and noise, and the 2014 demonstration was primarily an acquisition/reconstruction simulator [85]. MRXCAT2.0 adds statistical LV shape, biophysical healthy/infarct/dilated/hypertrophic function and generated tissue texture, exposing known simulated function for method testing [15, 86]. Both are generators with exemplar outputs, so neither has a fixed number of independent participants.

Large multi-centre resources such as CMRxRecon2025 remain useful for reconstruction robustness and disease representation. Its official description covers multiple centres, scanners, disease groups and CMR contrasts, and it does not establish tagging/DENSE motion labels or a fully paired cine–LGE cohort [87]. The same limit applies to segmentation collections and the Data Science Bowl: containing cine images does not make a collection a regional-motion reference. Detailed access, pairing, scale and uncertainty fields are provided in Supplementary Table S2 and the editable resource table.

Table 4. Complementary validation resources for cine-derived motion and strain. Counts denote the resource unit stated; they are not counts of independent test participants or complete paired analyses.

Resource	Nature and scale	Available pairing/reference	What it can evaluate	Access and principal boundary
STRAUS [75, 81]	Synthetic; 18 virtual cases from 3 templates × 6 functional states	3D cine, three tagged MR sequences and ultrasound with framewise VTK meshes	Dense correspondence, engineering strain, sampling and multimodal consistency in the simulated domain	Public, no login reported; only three source anatomies and imposed electromechanics/appearance
Stanford Cardiac Performance MRI [82]	Human; page: 51 healthy; DOI metadata: 55 total including 4 aortic stenosis	Whole-LV bSSFP cine; DENSE at matched locations; mid-LV tagging and other contrasts	In-vivo cross-sequence displacement/strain after explicit processing and alignment	Public repository, ODbL 1.0; completeness and usable pair count not yet file-verified
CMAC [83, 84]	Dynamic phantom and 15 healthy volunteers	Cine SSFP, 3D tagging and 3DUS; 12 manually tracked tagging landmarks	Sparse point correspondence and cross-modality benchmarking	Open access with attribution; sparse healthy reference, median inter-observer variability 0.84 mm
MRXCAT [85]	XCAT-based numerical generator, not a fixed human cohort	Prescribed anatomy/motion with cine/perfusion signal, coils, noise and sampling	Acquisition and reconstruction tests with known simulated motion	Code by non-commercial request; realism bounded by XCAT and signal assumptions
MRXCAT2.0 [15, 86]	Generator for healthy, infarcted, dilated and hypertrophic LV cases	Shape model, biophysical simulation, texture and known functional values	Capacity and estimation error across generated anatomy and pathophysiology	Repository cited by the paper; live contents not verified; generated instances are not independent clinical participants
CMRxRecon 2025 [87]	Multicentre, multivendor reconstruction resource with disease labels	Multiple CMR contrasts and reconstruction and quality-control material	Reconstruction robustness and domain shift	Challenge access terms; no declared dense motion truth or universal cine–LGE pairing

6.6 Reconstruction evidence is relevant and endpoint-specific

Reconstruction changes the images supplied to a downstream estimator, and primary studies show both agreement and residual bias. REGAIN was trained on 1616 patients and tested prospectively in 181 participants; its paired breath-hold comparison used 94 patients and 36 healthy participants, while global radial, circumferential and longitudinal feature-tracking strain was available in 49 participants. Mean global-strain differences versus a separately acquired GRAPPA cine were near zero, while the 95% limits of agreement remained −7% to 6% for radial, −2% to 3% for circumferential and −3% to 3% for longitudinal strain [19]. CineVN compared prospectively accelerated acquisitions with conventional GRAPPA in 18 healthy volunteers. On individual mid-ventricular slices, CineVN had the smallest bias and narrowest limits among the tested accelerated reconstructions, while all accelerated protocols significantly underestimated peak radial strain and overestimated peak circumferential strain; its 46-patient demonstration lacked a conventional reference and reported slice exclusions after segmentation failure [88]. These are measured reconstruction-plus-analysis results. Whether deep learning removes or preserves a focal abnormality is a separate question that they do not address.

Other designs bound the inference differently. In ten healthy volunteers, model-based deep-learning reconstruction reduced the dispersion of global circumferential strain differences from the fully sampled comparator relative to compressed sensing at reduction factors 6 and 8, while circumferential bias and

substantial radial underestimation remained and no regional lesion was tested [89]. In 19 participants (13 patients with coronary artery disease and six healthy controls) with 51 compressed-sensing acquisitions, retrospective temporal interpolation changed peak longitudinal and circumferential strain while ventricular volumes and ejection fraction stayed similar [90]; stable volumes do not guarantee stable strain. DENT was developed from 5840 patients across nine centres and compared interpolated with acquired cine phases, including prospective image-quality comparisons in 49 patients and 12 healthy participants. It reported image-domain interpolation error and reader assessment; myocardial strain and focal-abnormality recovery were outside its evaluation, and even its deformation vectors were not validated as motion estimates [91]. Reconstruction and motion estimation should therefore be evaluated jointly when the claim is regional, and any proposed mechanism (temporal smoothing, sharpening, alias suppression or hallucination) remains a hypothesis until tested against an attribute-appropriate reference.

7 Reporting and interpretation in practice

The consolidated reporting template is provided in Supplementary Note 5. Its essential logic is to identify the acquisition and reconstruction, estimator/version and interaction, complete strain estimand, statistical unit, validation target, failure policy and tested domain in the same record. For a focal claim, magnitude, location, extent and timing should be reported when each is relevant to the intended use, and an attribute that was not established here should keep that label.

The same record serves two audiences. For clinical readers, a regional strain map should be read together with the cine images, contours, tissue characterisation and global function. For technical readers, improving Dice, smoothness or Jacobian regularity is worthwhile, and the improvement should be connected to the regional quantity that motivates the method. For both, a negative result should examine possible contributions from insufficient input cues, inadequate representation, failed estimation, mismatched strain definition, reference uncertainty or lack of clinical association.

Software status needs the same discipline. Software clearance and commercial availability do not substitute for analytical validation of a specific regional output, and the existence of reference ranges does not make values interchangeable across products. Serial assessment is safest when acquisition, software, version and analysis conventions remain stable, and any change should be bridged empirically.

8 Limitations of the evidence and of this review

The evidence base is heterogeneous. Many studies are single-centre or use modest samples, healthy volunteers or a restricted disease spectrum. Paired DENSE/tagging studies face acquisition burden and alignment error. Commercial algorithms are not always described sufficiently to reproduce the exact mapping. Studies often report global or segmental peaks without the full field, and regional analyses may treat segments as independent. Publication bias may favour successful methods and well-behaved images.

This review has corresponding limits. It is a targeted narrative synthesis without duplicate screening, formal risk-of-bias scoring or pooled effects. Original exports, individual screening decisions and citation-expansion rounds from the historical search were not available for re-audit, so its counts and stopping rule remain inherited reports. October additions are separate from the September cutoff. Some recent method entries were inspected only at abstract or metadata depth; their summaries cannot establish that an unmentioned experiment was absent. The updated ledger distinguishes current primary-source checks, inherited details and unresolved access, and the literature map is not a denominator for field-wide prevalence. Product and regulatory details are outside the central synthesis because versions and indications change over time.

The proposed attribute-based framework is an organising approach; it is neither an established regulatory

standard nor a new validated biomarker property. Tolerances must be justified for each intended use. No review can determine from published aggregate results whether a particular patient's regional map is correct.

9 Conclusions

Routine cine CMR supports useful regional measurements, provided the claim is tied to the measured quantity and validation setting. Paired studies offer the clearest support for selected circumferential-strain endpoints at whole-slice, global and AHA-segment scales. StrainNet improved end-systolic cine–DENSE agreement over its tested conventional feature-tracking pipeline, while residual limits of agreement and internal validation constrain the inference. Repeatability, scar discrimination and prognosis contribute separate evidence about measurement precision and clinical relevance.

A claim about a focal abnormality requires evidence at that scale. DeepStrain on MRXCAT2.0 illustrates the distinction: component-specific whole-slice errors were reported in a prescribed-scar setting, but those averages do not quantify recovery inside the scar. Neither a lower mean circumferential error nor a clinically informative label establishes accurate lesion magnitude, boundary or timing. Apparent disagreement across studies can reflect software, sampling, strain convention, statistical unit and reference uncertainty, as well as estimation error; controlled comparisons are needed before attributing it to information loss or regularisation.

The practical requirement is a measurement contract linking acquisition, estimator, population, spatial unit, strain definition, time point, comparator and tolerance. Known-motion tests can probe representation and recovery; paired cine–DENSE or tagging comparisons can assess in-vivo agreement; tissue and outcome studies can test clinical relevance. Combining these complementary designs with participant-aware statistics offers a way to determine which abnormality attributes routine cine preserves and which remain unresolved, without asking a single validation target to establish them all.

Declarations

List of abbreviations

ACDC, Automated Cardiac Diagnosis Challenge; AHA, American Heart Association; AUC, area under the receiver operating characteristic curve; bSSFP, balanced steady-state free precession; CMAC, Cardiac Motion Analysis Challenge; CMR, cardiovascular magnetic resonance; CTEPH, chronic thromboembolic pulmonary hypertension; DENSE, displacement encoding with stimulated echoes; DLSS, deep-learning synthetic strain; Ecc, Err, Ell, circumferential, radial and longitudinal strain; EF, ejection fraction; EPE, end-point error; FT, feature tracking; ICC, intraclass correlation coefficient; LGE, late gadolinium enhancement; LS, CS, RS, longitudinal, circumferential and radial strain as labelled in the cited product comparison; LV, left ventricle or left-ventricular; MPN, Motion Pyramid Networks; ROC, receiver operating characteristic; RV, right ventricle or right-ventricular; SD, standard deviation; SNR, signal-to-noise ratio.

Ethics approval and consent to participate

Not applicable. This review did not collect new participant data.

Consent for publication

Not applicable to the review content. The corresponding author must confirm that all author and institutional details are approved before submission.

Availability of data and materials

No new participant dataset was created; the review uses published sources. The manuscript and editable figure sources, the surviving historical query log and the extraction notes are available from the corresponding author on reasonable request. Original historical search exports and record-level screening decisions are no longer available.

Competing interests

Author confirmation required before submission. This draft does not certify the absence or presence of competing interests.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI Codex and Claude Code (Anthropic) were used during preparation of this manuscript for search support, evidence organisation, drafting, revision, LaTeX preparation, deterministic consistency checks and programmatic figure generation. On 11 October 2026, OpenAI Codex was used to reconcile three AI-assisted review rounds of the draft with selected primary-source passages and with the revised figure package, and Claude Code was then used for a further scientific review and language revision whose edits are logged in the author-review package. AI-assisted review of intermediate drafts does not constitute independent human or clinical review. Figures were edited as Python/SVG sources and rendered with headless Chromium; the manuscript was compiled with pdfLaTeX and BibTeX. No generative image model, raw cardiac images or new experimental data were used. The corresponding author must verify the scientific content, sources, figures, process statements and declarations, confirm the journal's current policy on AI assistance and accept responsibility before submission.

Figure provenance and permissions

The figures combine conceptual diagrams, analytic illustrations and redraws of published numerical results. Figure 1 redraws reported values from MRXCAT2.0, Militaru, Cao, StrainNet, REGAIN, Strain-8, Kihlberg, Bello and DLSS; Figure 4 redraws MRXCAT2.0 values. Figure 3 maps selected source records and contains no measured patient data. Figures 2 and 5, Supplementary Figures S1–S3 and the graphical abstract contain conceptual or analytic schematics. No patient image or third-party figure artwork is copied. Editable figure sources and their numerical inputs accompany the figures in the author-review package. Scientific review of the figures, source acknowledgement, rights assessment and current journal-policy interpretation remain the author's responsibility and are pending.

References

[1] Schulz-Menger J, Bluemke DA, Bremerich J, Flamm SD, Fogel MA, Friedrich MG, et al. Standardized Image Interpretation and Post-Processing in Cardiovascular Magnetic Resonance—2020 Update. J Cardiovasc Magn Reson. 2020;22(1):19. doi:10.1186/s12968-020-00610-6.

[2] Hundley WG, Bluemke DA, Bogaert J, Flamm SD, Fontana M, Friedrich MG, et al. Society for Cardiovascular Magnetic Resonance (SCMR) Guidelines for Reporting Cardiovascular Magnetic Resonance Examinations. J Cardiovasc Magn Reson. 2022;24(1):29. doi:10.1186/s12968-021-00827-z.

[3] Buss SJ, Krautz B, Hofmann N, Sander Y, Rust L, Giusca S, et al. Prediction of Functional Recovery by Cardiac Magnetic Resonance Feature Tracking Imaging in First Time ST-Elevation Myocardial Infarction: Comparison to Infarct Size and Transmurality by Late Gadolinium Enhancement. Int J Cardiol. 2015;183:162-70. doi:10.1016/j.ijcard.2015.01.022.

[4] Reindl M, Tiller C, Holzknecht M, Lechner I, Eisner D, Riepl L, et al. Global Longitudinal Strain by Feature Tracking for Optimized Prediction of Adverse Remodeling after ST-Elevation Myocardial Infarction. Clin Res Cardiol. 2021;110(1):61-71. doi:10.1007/s00392-020-01649-2.

[5] Chadalavada S, Fung K, Rauseo E, Lee AM, Khanji MY, Amir-Khalili A, et al. Myocardial Strain Measured by Cardiac Magnetic Resonance Predicts Cardiovascular Morbidity and Death. J Am Coll Cardiol. 2024;84(7):648-59. doi:10.1016/j.jacc.2024.05.050.

[6] Mangion K, Carrick D, Carberry J, Mahrous A, McComb C, Oldroyd KG, et al. Circumferential Strain Predicts Major Adverse Cardiovascular Events Following an Acute ST-Segment–Elevation Myocardial Infarction. Radiology. 2019;290(2):329-37. doi:10.1148/radiol.2018181253.

[7] Raunig DL, McShane LM, Pennello G, Gatsonis C, Carson PL, Voyvodic JT, et al. Quantitative Imaging Biomarkers: A Review of Statistical Methods for Technical Performance Assessment. Stat Methods Med Res. 2015;24(1):27-67. doi:10.1177/0962280214537344.

[8] Puntmann VO, Valbuena S, Hinojar R, Petersen SE, Greenwood JP, Kramer CM, et al. Society for Cardiovascular Magnetic Resonance (SCMR) Expert Consensus for CMR Imaging Endpoints in Clinical Research: Part I—Analytical Validation and Clinical Qualification. J Cardiovasc Magn Reson. 2018;20(1):67. doi:10.1186/s12968-018-0484-5.

[9] Amzulescu MS, De Craene M, Langet H, Pasquet A, Vancraeynest D, Pouleur AC, et al. Myocardial Strain Imaging: Review of General Principles, Validation, and Sources of Discrepancies. Eur Heart J Cardiovasc Imaging. 2019;20(6):605-19. doi:10.1093/ehjci/jez041.

[10] Smiseth OA, Rider O, Cvijic M, Valkovič L, Remme EW, Voigt JU. Myocardial Strain Imaging: Theory, Current Practice, and the Future. JACC Cardiovasc Imaging. 2025;18(3):340-81. doi:10.1016/j.jcmg.2024.07.011.

[11] Monfared M, Kakavand B, Taebi A. Deep Learning for Cardiac Wall Motion Analysis: A Review of Methods, Challenges, and Clinical Applications. Ann Biomed Eng. 2026. Published online 29 April 2026. doi:10.1007/s10439-026-04147-0.

[12] Hussein TM, Pourmorteza A, Oshinski JN. Myocardial Strain Evaluation from Cardiovascular MRI and CT: Physiological Principles and Clinical Perspectives. Br J Radiol. 2026;99(1186):2004-19. doi:10.1093/bjr/tqag192.

[13] Pedrizzetti G, Claus P, Kilner PJ, Nagel E. Principles of Cardiovascular Magnetic Resonance Feature Tracking and Echocardiographic Speckle Tracking for Informed Clinical Use. J Cardiovasc Magn Reson. 2016;18(1):51. doi:10.1186/s12968-016-0269-7.

[14] Duchateau N, King AP, De Craene M. Machine Learning Approaches for Myocardial Motion and Deformation Analysis. Front Cardiovasc Med. 2020;6:190. doi:10.3389/fcvm.2019.00190.

[15] Buoso S, Joyce T, Schulthess N, Kozerke S. MRXCAT2.0: Synthesis of Realistic Numerical Phantoms by Combining Left-Ventricular Shape Learning, Biophysical Simulations and Tissue Texture Generation. J Cardiovasc Magn Reson. 2023;25(1):25. doi:10.1186/s12968-023-00934-z.

[16] Militaru S, Panovsky R, Hanet V, Amzulescu MS, Langet H, Pisciotti MM, et al. Multivendor Comparison of Global and Regional 2D Cardiovascular Magnetic Resonance Feature Tracking Strains vs Tissue Tagging at 3T. J Cardiovasc Magn Reson. 2021;23(1):54. doi:10.1186/s12968-021-00742-3.

[17] Cao JJ, Ngai N, Duncanson L, Cheng J, Gliganic K, Chen Q. A Comparison of Both DENSE and Feature Tracking Techniques with Tagging for the Cardiovascular Magnetic Resonance Assessment of Myocardial Strain. J Cardiovasc Magn Reson. 2018;20(1):26. doi:10.1186/s12968-018-0448-9.

[18] Wang Y, Sun C, Ghadimi S, Auger DC, Croisille P, Viallon M, et al. StrainNet: Improved Myocardial Strain Analysis of Cine MRI by Deep Learning from DENSE. Radiol Cardiothorac Imaging. 2023;5(3):e220196. doi:10.1148/ryct.220196.

[19] Yoon S, Nakamori S, Amyar A, Assana S, Cirillo J, Morales MA, et al. Accelerated Cardiac MRI Cine with Use of Resolution Enhancement Generative Adversarial Inline Neural Network. Radiology. 2023;307(5):e222878. doi:10.1148/radiol.222878.

[20] Bell A, Okafor J, Yazdani M, Franks R, Draper J, Campbell B, et al. The Strain-8 Study: A Multimodal Scan–Rescan Assessment of Myocardial Strain Repeatability. Eur Heart J Imaging Methods Pract. 2025;3(4):qyaf144. doi:10.1093/ehjimp/qyaf144.

[21] Kihlberg J, Gupta V, Haraldsson H, Sigfridsson A, Sarvari SI, Ebbers T, et al. Clinical Validation of Three Cardiovascular Magnetic Resonance Techniques to Measure Strain and Torsion in Patients with Suspected Coronary Artery Disease. J Cardiovasc Magn Reson. 2020;22(1):83. doi:10.1186/s12968-020-00684-2.

[22] Bello GA, Dawes TJW, Duan J, Biffi C, de Marvao A, Howard LSGE, et al. Deep-Learning Cardiac Motion Analysis for Human Survival Prediction. Nat Mach Intell. 2019;1(2):95-104. doi:10.1038/s42256-019-0019-2.

[23] Masutani EM, Chandrupatla RS, Wang S, Zocchi C, Hahn LD, Horowitz M, et al. Deep Learning Synthetic Strain: Quantitative Assessment of Regional Myocardial Wall Motion at MRI. Radiol Cardiothorac Imaging. 2023;5(3):e220202. doi:10.1148/ryct.220202.

[24] Voigt JU, Pedrizzetti G, Lysyansky P, Marwick TH, Houle H, Baumann R, et al. Definitions for a Common Standard for 2D Speckle Tracking Echocardiography: Consensus Document of the EACVI/ASE/Industry Task Force to Standardize Deformation Imaging. Eur Heart J Cardiovasc Imaging. 2015;16(1):1-11. doi:10.1093/ehjci/jeu184.

[25] Wehner GJ, Jing L, Haggerty CM, Suever JD, Chen J, Hamlet SM, et al. Comparison of Left Ventricular Strains and Torsion Derived from Feature Tracking and DENSE CMR. J Cardiovasc Magn Reson. 2018;20(1):63. doi:10.1186/s12968-018-0485-4.

[26] Metz CT, Klein S, Schaap M, van Walsum T, Niessen WJ. Nonrigid Registration of Dynamic Medical Imaging Data Using nD+t B-Splines and a Groupwise Optimization Approach. Med Image Anal. 2011;15(2):238-49. doi:10.1016/j.media.2010.10.003.

[27] Chen H, Gao J, Chen Z, Gao C, Huo S, Jiang M, et al. Improve Myocardial Strain Estimation Based on Deformable Groupwise Registration with a Locally Low-Rank Dissimilarity Metric. BMC Med

Imaging. 2024;24(1):330. doi:10.1186/s12880-024-01519-7.

[28] Lim C, Blaszczyk E, Riazy L, Wiesemann S, Schüler J, von Knobelsdorff-Brenkenhoff F, et al. Quantification of Myocardial Strain Assessed by Cardiovascular Magnetic Resonance Feature Tracking in Healthy Subjects—Influence of Segmentation and Analysis Software. Eur Radiol. 2021;31(6):3962-72. doi:10.1007/s00330-020-07539-5.

[29] Maret E, Todt T, Brudin L, Nylander E, Swahn E, Ohlsson JL, et al. Functional measurements based on feature tracking of cine magnetic resonance images identify left ventricular segments with myocardial scar. Cardiovascular Ultrasound. 2009;7(1):53. doi:10.1186/1476-7120-7-53.

[30] Hor KN, Gottliebson WM, Carson C, Wash E, Cnota J, Fleck R, et al. Comparison of Magnetic Resonance Feature Tracking for Strain Calculation With Harmonic Phase Imaging Analysis. JACC: Cardiovascular Imaging. 2010;3(2):144-51. doi:10.1016/j.jcmg.2009.11.006.

[31] Augustine D, Lewandowski AJ, Lazdam M, Rai A, Francis J, Myerson S, et al. Global and regional left ventricular myocardial deformation measures by magnetic resonance feature tracking in healthy volunteers: comparison with tagging and relevance of gender. Journal of Cardiovascular Magnetic Resonance. 2013;15(1):8. doi:10.1186/1532-429X-15-8.

[32] Vigneault DM, Xie W, Bluemke DA, Noble JA. Feature Tracking Cardiac Magnetic Resonance via Deep Learning and Spline Optimization. In: Functional Imaging and Modeling of the Heart. Springer International Publishing; 2017. p. 183-94. doi:10.1007/978-3-319-59448-4_18.

[33] Puyol-Anton E, Ruijsink B, Bai W, Langet H, De Craene M, Schnabel JA, et al. Fully Automated Myocardial Strain Estimation from Cine MRI Using Convolutional Neural Networks. In: 2018 IEEE 15th International Symposium on Biomedical Imaging; 2018. p. 1139-43. doi:10.1109/ISBI.2018.8363772.

[34] Rueckert D, Sonoda LI, Hayes C, Hill DLG, Leach MO, Hawkes DJ. Nonrigid Registration Using Free-Form Deformations: Application to Breast MR Images. IEEE Trans Med Imaging. 1999;18(8):712-21. doi:10.1109/42.796284.

[35] Wang L, Clarysse P, Liu Z, Gao B, Liu W, Croisille P, et al. A gradient-based optical-flow cardiac motion estimation method for cine and tagged MR images. Medical Image Analysis. 2019;57:136-48. doi:10.1016/j.media.2019.06.016.

[36] Elnakib A, Beache GM, Gimel'farb G, El-Baz A. Intramyocardial strain estimation from cardiac cine MRI. International Journal of Computer Assisted Radiology and Surgery. 2015;10(8):1299-312. doi:10.1007/s11548-014-1137-2.

[37] Tuyisenge V, Sarry L, Corpetti T, Innorta-Coupez E, Ouchchane L, Cassagnes L. Estimation of Myocardial Strain and Contraction Phase From Cine MRI Using Variational Data Assimilation. IEEE Transactions on Medical Imaging. 2016;35(2):442-55. doi:10.1109/TMI.2015.2478117.

[38] Chen P, Chen X, Chen EZ, Yu H, Chen T, Sun S. Anatomy-Aware Cardiac Motion Estimation. In: Machine Learning in Medical Imaging. Springer; 2020. p. 150-9. doi:10.1007/978-3-030-59861-7_16.

[39] Yu H, Chen X, Shi H, Chen T, Huang TS, Sun S. Motion Pyramid Networks for Accurate and Efficient Cardiac Motion Estimation. In: Medical Image Computing and Computer Assisted Intervention – MICCAI 2020. Springer; 2020. p. 436-46. doi:10.1007/978-3-030-59725-2_42.

[40] Yu H, Sun S, Yu H, Chen X, Shi H, Huang TS, et al. FOAL: Fast Online Adaptive Learning for Cardiac Motion Estimation. In: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition; 2020. p. 4312-22. doi:10.1109/CVPR42600.2020.00437.

[41] Balakrishnan G, Zhao A, Sabuncu MR, Guttag J, Dalca AV. VoxelMorph: A Learning Framework for Deformable Medical Image Registration. IEEE Trans Med Imaging. 2019;38(8):1788-800. doi:10.1109/TMI.2019.2897538.

[42] Chen J, Frey EC, He Y, Segars WP, Li Y, Du Y. TransMorph: Transformer for Unsupervised Medical Image Registration. Med Image Anal. 2022;82:102615. doi:10.1016/j.media.2022.102615.

[43] Jia X, Bartlett J, Chen W, Song S, Zhang T, Cheng X, et al. Fourier-Net: Fast Image Registration with Band-Limited Deformation. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 37; 2023. p. 1015-23. Issue 1. doi:10.1609/aaai.v37i1.25182.

[44] Jia X, Thorley A, Gomez A, Lu W, Kotecha D, Frangi AF, et al. Fourier-Net+: Band-Limited Spatial Representation for Efficient Medical Image Registration. IEEE Trans Neural Netw Learn Syst. 2026;37(10):5712-26. Published online 19 May 2026. doi:10.1109/TNNLS.2026.3688245.

[45] Qin C, Bai W, Schlemper J, Petersen SE, Piechnik SK, Neubauer S, et al. Joint Learning of Motion Estimation and Segmentation for Cardiac MR Image Sequences. In: Medical Image Computing and Computer Assisted Intervention – MICCAI 2018. Springer; 2018. p. 472-80. doi:10.1007/978-3-030-00934-2_53.

[46] Zheng Q, Delingette H, Ayache N. Explainable Cardiac Pathology Classification on Cine MRI with Motion Characterization by Semi-Supervised Learning of Apparent Flow. Med Image Anal. 2019;56:80-95. doi:10.1016/j.media.2019.06.001.

[47] Portal N, Dietenbeck T, Khan S, Nguyen V, Prigent M, Zarai M, et al. Semi-Supervised Motion Flow and Myocardial Strain Estimation in Cardiac Videos Using Distance Maps and Memory Networks. Comput Biol Med. 2025;196:110739. doi:10.1016/j.compbiomed.2025.110739.

[48] Meng Q, Qin C, Bai W, Liu T, de Marvao A, O'Regan DP, et al. MulViMotion: Shape-Aware 3D Myocardial Motion Tracking from Multi-View Cardiac MRI. IEEE Trans Med Imaging. 2022;41(8):1961-74. doi:10.1109/TMI.2022.3154599.

[49] Meng Q, Bai W, O'Regan DP, Rueckert D. DeepMesh: Mesh-Based Cardiac Motion Tracking Using Deep Learning. IEEE Trans Med Imaging. 2024;43(4):1489-500. doi:10.1109/TMI.2023.3340118.

[50] Movahed RA, Rezaee A, Zakeri A, Berry C, Ho ESL, Gooya A. CardioMorphNet: Cardiac Motion Prediction Using a Shape-Guided Bayesian Recurrent Deep Network. Med Image Anal. 2026;113:104149. doi:10.1016/j.media.2026.104149.

[51] Qiao X, Yu J, Wang H, Ding W, Zhang R, Zhu Z, et al. STCMT-Net: A Spatiotemporal Consistency Motion Transfer Network for Enhancing Cardiac Motion Estimation. Med Image Anal. 2026;109:103926. doi:10.1016/j.media.2025.103926.

[52] Lu J, Jin R, Wang M, Song E, Ma G. A bidirectional registration neural network for cardiac motion tracking using cine MRI images. Computers in Biology and Medicine. 2023;160:107001. doi:10.1016/j.compbiomed.2023.107001.

[53] Bi N, Zakeri A, Xia Y, Cheng N, Frangi AF, Gooya A. SegMorph: Concurrent Motion Estimation and Segmentation for Cardiac MRI Sequences. IEEE Transactions on Medical Imaging. 2025;44(9):3515-28. doi:10.1109/TMI.2024.3435000.

[54] Jia S, Yang Y, Chang Q. IG-BiReg: implicit-guided bidirectional residual registration framework for cardiac cine MRI images. Physics in Medicine & Biology. 2026. Online first, 5 October 2026; volume and pages not supplied in the retrieved metadata. doi:10.1088/1361-6560/aeb060.

[55] Yin J, Cheng X, Duan J, Pang Y, O'Regan D, Reynaud H, et al.. MCM: Mamba-Based Cardiac

Motion Tracking Using Sequential Images in MRI; 2025. MICCAI 2025 Reconstruction and Imaging Motion Estimation workshop paper; author version. arXiv:2507.17678 [eess.IV]. doi:10.48550/arXiv.2507.17678.

[56] Ruan D, Zhou K, Zhu C, Yu C, Gao Z, Jiang M, et al. Cardiac Motion Tracking with Bidirectional Latent Neural ODEs from Cine Cardiac MRI. Comput Med Imaging Graph. 2026;134:102797. doi:10.1016/j.compmedimag.2026.102797.

[57] Fu Y, Bai W, Yi W, Manisty C, Bhuva AN, Treibel TA, et al. Development and Validation of a Versatile Foundation Model for Cine Cardiac Magnetic Resonance Image Analysis. Commun Med (Lond). 2026;6(1):408. doi:10.1038/s43856-026-01636-0.

[58] Jacob AJ, Borgohain I, Chitiboi T, Sharma P, Comaniciu D, Rueckert D. Towards a Cardiovascular Magnetic Resonance Foundation Model for Multi-Task Cardiac Image Analysis. J Cardiovasc Magn Reson. 2025;27(2):101967. doi:10.1016/j.jocmr.2025.101967.

[59] Qin C, Wang S, Chen C, Bai W, Rueckert D. Generative Myocardial Motion Tracking via Latent Space Exploration with Biomechanics-Informed Prior. Med Image Anal. 2023;83:102682. doi:10.1016/j.media.2022.102682.

[60] Arratia López P, Mella H, Uribe S, Hurtado DE, Sahli Costabal F. WarpPINN: Cine-MR Image Registration with Physics-Informed Neural Networks. Med Image Anal. 2023;89:102925. doi:10.1016/j.media.2023.102925.

[61] Álvarez-Barrientos F, Banduc T, Sirven I, Sahli Costabal F. WarpPINN-fibers: Improved Cardiac Strain Estimation from Cine-MR with Physics-Informed Neural Networks. Engineering with Computers. 2026;42(2):58. doi:10.1007/s00366-026-02295-4.

[62] Liu H, Qin X, Liu Z, Jin Y, Jiang H, Gao Y, et al. Unlocking 2D/3D+T Myocardial Mechanics from Cine MRI: A Mechanically Regularized Space-Time Finite Element Correlation Framework. Med Image Anal. 2026;109:103944. doi:10.1016/j.media.2026.103944.

[63] Ye M, Kanski M, Yang D, Chang Q, Yan Z, Huang Q, et al. DeepTag: An Unsupervised Deep Learning Method for Motion Tracking on Cardiac Tagging Magnetic Resonance Images. In: 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition; 2021. p. 7257-67. doi:10.1109/CVPR46437.2021.00718.

[64] Buoso S, Stoeck CT, Kozerke S. Automatic Analysis of Three-Dimensional Cardiac Tagged Magnetic Resonance Images Using Neural Networks Trained on Synthetic Data. J Cardiovasc Magn Reson. 2025;27:101869. doi:10.1016/j.jocmr.2025.101869.

[65] Morales MA, van den Boomen M, Nguyen C, Kalpathy-Cramer J, Rosen BR, Stultz CM, et al. DeepStrain: A Deep Learning Workflow for the Automated Characterization of Cardiac Mechanics. Front Cardiovasc Med. 2021;8:730316. doi:10.3389/fcvm.2021.730316.

[66] Morales MA, Cirillo J, Nakata K, Kucukseymen S, Ngo LH, Izquierdo-Garcia D, et al. Comparison of DeepStrain and Feature Tracking for Cardiac MRI Strain Analysis. J Magn Reson Imaging. 2023;57(5):1507-15. doi:10.1002/jmri.28374.

[67] Lei P, Xing J, Fang F, Epstein FH, Zhang M. DENSE-Guided Deep Motion Networks Accounted by Large Rotations to Improve Myocardial Strain Analysis from Routine Cine MRI. In: Proc IEEE Int Symp Biomed Imaging (ISBI); 2025. p. 1-5. doi:10.1109/isbi60581.2025.10980688.

[68] An D, Nencka A, Clarysse P, Croisille P, Bergom C, Ibrahim ES. MyoNet: Deep Learning-Based Myocardial Strain Quantification from Cine Cardiac MRI. Bioengineering (Basel). 2026;13(3):310. doi:10.3390/bioengineering13030310.

[69] Xu C, Xu L, Gao Z, Zhao S, Zhang H, Zhang Y, et al. Direct Delineation of Myocardial Infarction without Contrast Agents Using a Joint Motion Feature Learning Architecture. Med Image Anal. 2018;50:82-94. doi:10.1016/j.media.2018.09.001.

[70] Koehler S, Kuhm J, Huffaker T, Young D, Tandon A, André F, et al. Deep Learning-Based Aligned Strain from Cine Cardiac MRI for Detection of Fibrotic Myocardial Tissue in Patients with Duchenne Muscular Dystrophy. Radiol Artif Intell. 2025;7(3):e240303. doi:10.1148/ryai.240303.

[71] Adams DM, Boubertakh R, Miquel ME. Effects of Spatial and Temporal Resolution on Cardiovascular Magnetic Resonance Feature Tracking Measurements Using a Simple Realistic Numerical Phantom. Br J Radiol. 2023;96(1143):20220233. doi:10.1259/bjr.20220233.

[72] Mukherjee T, Usman M, Mehdi RR, Mendiola E, Ohayon J, Lindquist D, et al. In-Silico Heart Model Phantom to Validate Cardiac Strain Imaging. Comput Biol Med. 2024;181:109065. doi:10.1016/j.compbiomed.2024.109065.

[73] Barbaroux H, Loecher M, Brackenier Y, Kunze KP, Neji R, Pennell DJ, et al. DENSE-SIM: A Modular Pipeline for the Evaluation of Cine Displacement Encoding with Stimulated Echoes Images with Sub-Voxel Ground-Truth Strain. J Cardiovasc Magn Reson. 2025;27(1):101866. doi:10.1016/j.jocmr.2025.101866.

[74] Yeon SB, Reichek N, Tallant BA, Lima JAC, Calhoun LP, Clark NR, et al. Validation of In Vivo Myocardial Strain Measurement by Magnetic Resonance Tagging with Sonomicrometry. J Am Coll Cardiol. 2001;38(2):555-61. doi:10.1016/s0735-1097(01)01397-3.

[75] Zhou Y, Giffard-Roisin S, De Craene M, Camarasu-Pop S, D'hooge J, Alessandrini M, et al. A Framework for the Generation of Realistic Synthetic Cardiac Ultrasound and Magnetic Resonance Imaging Sequences from the Same Virtual Patients. IEEE Trans Med Imaging. 2018;37(3):741-54. doi:10.1109/TMI.2017.2708159.

[76] Gao H, Allan A, McComb C, Luo X, Berry C. Left Ventricular Strain and Its Pattern Estimated from Cine CMR and Validation with DENSE. Phys Med Biol. 2014;59(13):3637-56. doi:10.1088/0031-9155/59/13/3637.

[77] Brandt Y, Lubrecht JM, Adriaans BP, Aben JP, Gerretsen SC, Ghossein-Doha C, et al. Quantification of Left Ventricular Myocardial Strain: Comparison between MRI Tagging, MRI Feature Tracking, and Ultrasound Speckle Tracking. NMR Biomed. 2024;37(9):e5164. doi:10.1002/nbm.5164.

[78] Naqizadeh Jahromi M, Menna Costa R, Delavald Marques A, Wellner L, Hannum AJ, Liu ZQ, et al. Automated cardiac strain estimation from short-axis DENSE and Cine MR images. Biomech Model Mechanobiol. 2026;25(5):103. Published online 15 September 2026. Available from: https://doi.org/10.1007/s10237-026-02120-3. doi:10.1007/s10237-026-02120-3.

[79] Goetze C, Chen W, Doeblin P, Demir A, Wiesemann S, Hansmann J, et al. A Multicenter Study on the Comparability of Myocardial Strain Values Acquired with Different CMR Scanners and Analyzed with Different Post-Processing Software: Insights into the "Traveling Volunteers" Study. Int J Comput Assist Radiol Surg. 2026;21(2):319-31. doi:10.1007/s11548-025-03499-7.

[80] He J, Kong L, An D, Chen B, Zhao C, Li Z, et al. Prognostic Value of Segmental Strain After ST-Elevation Myocardial Infarction: Insights From the EARLY Assessment of MYOcardial Tissue Characteristics by Cardiac Magnetic Resonance (EARLY-MYO-CMR) Study. J Magn Reson Imaging. 2024;60(5):2002-17. doi:10.1002/jmri.29274.

[81] CREATIS Human Heart Project. Multimodality STRAUS; n.d. Public database; accessed 8 October 2026. Available from: https://humanheart-project.creatis.insa-lyon.fr/

multimodalityStraus.html.

[82] Cardiac Magnetic Research Group. Human MRI Data to Quantify Cardiac Performance. Stanford Digital Repository; 2026. DataCite metadata updated 27 July 2026; ODbL 1.0; accessed 8 October 2026. Available from: https://purl.stanford.edu/xt223hz2438. doi:10.25740/xt223hz2438.

[83] Tobon-Gomez C, De Craene M, McLeod K, Tautz L, Shi W, Hennemuth A, et al. Benchmarking Framework for Myocardial Tracking and Deformation Algorithms: An Open Access Database. Med Image Anal. 2013;17(6):632-48. doi:10.1016/j.media.2013.03.008.

[84] Cardiac Atlas Project. Cardiac Motion Analysis Challenge 2011; 2011. Accessed 8 October 2026. Available from: https://www.cardiacatlas.org/motion-tracking-2011-challenge/.

[85] Wissmann L, Santelli C, Segars WP, Kozerke S. MRXCAT: Realistic Numerical Phantoms for Cardiovascular Magnetic Resonance. J Cardiovasc Magn Reson. 2014;16(1):63. doi:10.1186/s12968-014-0063-3.

[86] Institute for Biomedical Engineering, ETH Zurich. MRXCAT 2.0 Source Repository; n.d. Repository linked by the primary paper; live contents and creation date not verified on 11 October 2026. Available from: https://gitlab.ethz.ch/ibt-cmr-public/mrxcat-2.0.

[87] CMRxRecon Organising Committee. CMRxRecon2025: Multi-Centre, Multi-Vendor and Multi-Disease Cardiovascular Magnetic Resonance Reconstruction Challenge; 2025. Official challenge site; modalities and release status checked 7 October 2026. Available from: https://cmrxrecon.github.io/2025/.

[88] Vornehm M, Wetzl J, Giese D, Fürnrohr F, Pang J, Chow K, et al. CineVN: Variational Network Reconstruction for Rapid Functional Cardiac Cine MRI. Magn Reson Med. 2025;93(1):138-50. Published online 27 August 2024. doi:10.1002/mrm.30260.

[89] Tsuneta S, Aono S, Kimura R, Kwon J, Fujima N, Ishizaka K, et al. The Impact of Model-Based Deep-Learning Reconstruction Compared with That of Compressed Sensing–Sensitivity Encoding on the Image Quality and Precision of Cine Cardiac MR in Evaluating Left-Ventricular Volume and Strain: A Study on Healthy Volunteers. Magn Reson Med Sci. 2025;24(4):mp.2024-0202. doi:10.2463/mrms.mp.2024-0202.

[90] Grob L, Schwerzmann Y, Kaiser D, Jung B, Schweizer T, Huettenmoser SP, et al. Retrospective temporal resolution interpolation alters myocardial strain quantification on compressed sensing cine CMR. The International Journal of Cardiovascular Imaging. 2025;41(3):591-602. doi:10.1007/s10554-025-03348-3.

[91] Morales MA, Ghanbari F, Nakamori S, Assana S, Amyar A, Yoon S, et al. Deformation-Encoding Deep Learning Transformer for High-Frame-Rate Cardiac Cine MRI. Radiol Cardiothorac Imaging. 2024;6(3):e230177. doi:10.1148/ryct.230177.